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A syngeneic spontaneous zebrafish model of *tp53*-deficient, EGFR^{vIII}, and PI3KCA^{H1047R}-driven glioblastoma reveals inhibitory roles for inflammation during tumor initiation and relapse *in vivo*

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Abstract

High-throughput vertebrate animal model systems for the study of patient-specific biology and new therapeutic approaches for aggressive brain tumors are currently lacking, and new approaches are urgently needed. Therefore, to build a patient-relevant *in vivo* model of human glioblastoma, we expressed common oncogenic variants including activated human EGFR^{vIII} and PI3KCA^{H1047R} under the control of the radial glial-specific promoter *her4.1* in syngeneic *tp53* loss-of-function mutant zebrafish. Robust tumor formation was observed prior to 45 days of life, and tumors had a gene expression signature similar to human glioblastoma of the mesenchymal subtype, with a strong inflammatory component. Within early stage tumor lesions, and in an *in vivo* and endogenous tumor microenvironment, we visualized infiltration of phagocytic cells, as well as internalization of tumor cells by *mpeg1.1*:EGFP+ microglia/macrophages, suggesting negative regulatory pressure by pro-inflammatory cell types on tumor growth at early stages of glioblastoma initiation. Furthermore, CRISPR/Cas9-mediated gene targeting of master inflammatory transcription factors *irf7* or *irf8* led to increased tumor formation in the primary context, while suppression of phagocyte activity led to enhanced tumor cell engraftment following transplantation into otherwise immune-competent zebrafish hosts. Altogether, we developed a genetically relevant model of aggressive human glioblastoma and harnessed the unique advantages of zebrafish including live imaging, high-throughput genetic and chemical manipulations to highlight important tumor suppressive roles for the innate immune system on glioblastoma initiation, with important future opportunities for therapeutic discovery and optimizations.

eLife assessment

This study presents a **valuable** syngeneic zebrafish model for studying glioblastoma and will be of interest to neuro-oncologists and cancer biologists. Using a feasible *in vivo* model to study the tumour microenvironment, cell/cell interaction, and immunity, the data are **compelling**, and opens up new lines of inquiries for future investigation on the impact of efferocytosis on tumor progression and cell of origin in this model as well as assessments of drug resistance mechanisms, using inhibitors to MAPK, Akt and/or mTOR pathway.

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Introduction

Creating faithful models and discovering tailored treatments for patients with aggressive brain tumors has resulted in many different experimental platforms, each with their own unique advantages (Gómez-Oliva et al., 2021 [↗](#); Haddad et al., 2021 [↗](#)). However, tractable vertebrate animal model systems for high-throughput study of tumor development in an intact and endogenous tumor microenvironment (TME) remain limited. New approaches that reflect patient genetics and physiology have the potential to aid in improving therapeutic strategies for individuals with malignant brain tumors like glioblastoma, which represents about 50% of all primary brain malignancies in adults (Ostrom et al., 2017 [↗](#); Torp et al., 2022 [↗](#)).

Despite intensive treatments including surgery, radiation, and chemotherapy (temozolomide, TMZ), most patients with glioblastoma eventually relapse and have a median survival rate of less than 15 months (Aldape et al., 2019 [↗](#)). Therapeutic resistance can partly be attributed to a poor understanding of underlying molecular mechanisms, as well as a significant level of heterogeneity between patients and within individual tumors. Molecular heterogeneity has been important for the classification of three major subtypes of glioblastoma including proneural, classical, and mesenchymal (Brennan et al., 2013 [↗](#); Torp et al., 2022 [↗](#); Verhaak et al., 2010 [↗](#); Wang et al., 2017 [↗](#)). However, recent evidence supporting co-existence of inter-converting glioblastoma cell states within individual patient tumors reveals less distinct subtype separations and phenotypic flexibilities that contribute to the aggressiveness and drug resistance of glioblastoma across molecular cohorts (Darmanis et al., 2017 [↗](#); Neftel et al., 2019 [↗](#); Patel et al., 2014 [↗](#)).

In general, 90% of glioblastoma tumors display alterations in core signaling factors involved in Receptor Tyrosine Kinase (RTK)/RAS/PI3K pathway activation (Brennan et al., 2013 [↗](#); McLendon et al., 2008 [↗](#)). While activating mutations in RAS proteins are rarely found in the clinic, loss of the negative regulator *NF1* is common, as well as amplifications and/or activating mutations in various RAS pathway activators including Epidermal Growth Factors Receptor (EGFR), which are found in >50% of glioblastomas (Brennan et al., 2013 [↗](#); Hoogstrate et al., 2022 [↗](#); McLendon et al., 2008 [↗](#)). Loss of the lipid phosphatase PTEN and/or activating mutations in the PI3K catalytic subunit PIK3CA are also commonly found, which altogether drive downstream activation of oncogenic RAS/MAPK and AKT/mTOR signaling, among other crucial pathways involved in growth and survival of glioblastoma (Brennan et al., 2013 [↗](#); McLendon et al., 2008 [↗](#)). Furthermore, the TP53 tumor suppressor pathway is altered in 84% of glioblastoma patients and 94% of cell lines, with TP53 loss implicated in proliferation, invasion, migration, and stemness (Brennan et al., 2013 [↗](#); McLendon et al., 2008 [↗](#); Zhang et al., 2018 [↗](#)). Importantly, single pathway mutations are generally insufficient to transform normal brain tissues, and multiple mutations are required for

glioma formation (Chen et al., 2018 [↗](#); Haddad et al., 2021 [↗](#); Holland et al., 2013 [↗](#); Takke et al., 1999 [↗](#)). Therefore, to recapitulate human glioblastoma in experimental models, multiple genetic events should be considered.

In addition to key genetic drivers, the TME is known to influence cellular flexibility in glioblastoma and consists of heterogeneous collections of resident brain, stroma, and immune cells, as well as cells recruited from the general circulation such as bone marrow-derived immune cells with known tumor suppressive and tumor promoting functions (Bikfalvi et al., 2023 [↗](#); Quail and Joyce, 2017 [↗](#)). Given emerging opportunities for harnessing the immune system for the treatment of human cancer, there is a growing focus on understanding innate and adaptive immune responses across different subtypes of human malignancies. However, like many other tumors, glioblastoma combines a lack of immunogenicity due to few mutations with a highly immunosuppressive tumor microenvironment. In addition to off-target effects of current frontline therapeutic strategies, both tumor and immune cells contribute to immune suppression in and surrounding the TME in glioblastoma (McGranahan et al., 2019 [↗](#); Sengupta et al., 2012 [↗](#)), which could explain the failure of immunotherapy-based clinical trials. Additionally, lymphocytes are frequently exhausted and dysfunctional and therefore inadequate at exerting an anti-tumor immune response, while tumor-associated myeloid cells are frequently reprogrammed by signaling from tumor cells and the TME to cell states that promote glioblastoma survival, growth, and invasion (Kennedy et al., 2013 [↗](#); McGranahan et al., 2019 [↗](#); Pearson et al., 2020 [↗](#)). Evidence also supports tumor-cell intrinsic mechanisms in response to immune cell attack leading to various evasion mechanisms, including upregulation of myeloid-associated gene expression programs and resistance to interferon signaling (Gangoso et al., 2021 [↗](#); Parmigiani et al., 2022 [↗](#)).

Complex intercellular communication in glioblastoma highlights the importance of faithful *in vivo* models. However, roles for the endogenous TME especially at early stages of tumor initiation, remain poorly understood. Furthermore, live visualization of heterogeneity and tumor cells within the TME is limited in non-transparent rodent genetic models and/or patient-derived tumor xenografted hosts. Therefore, here our goal was to develop a patient-relevant model of aggressive human glioma in an intact and immune competent system. We developed a novel spontaneous, syngeneic zebrafish model of glioblastoma, with high levels of inflammatory immune cell infiltration and anti-tumor associations between phagocytes in the TME and tumor cells, suggesting inhibitory roles for inter-cellular interactions during glioblastoma initiation and an effective *in vivo* platform for future biological discovery and drug testing for patients.

Results

Oncogenic MAPK/AKT pathway activation drives glial-derived tumor formation in syngeneic *tp53* mutant zebrafish

To generate a patient-relevant brain tumor model in zebrafish we used the zebrafish *her4.1* promoter to simultaneously over-express constitutively active human EGFR (EGFR^{VIII}) and PI3KCA (PI3KCA^{H1047R}) variants in neural progenitors and radial glia of syngeneic *tp53* loss-of-function mutant larvae (Ignatius et al., 2018 [↗](#); Takke et al., 1999 [↗](#); Than-Trong et al., 2020 [↗](#)). Co-injection of linearized *her4.1:EGFR^{VIII}* + *her4.1:PI3KCA^{H1047R}* + *her4.1:mScarlet* transgenes into syngeneic (CG1) *tp53*^{-/-} mutant embryos at the one-cell stage led to broad transient mScarlet expression for 5-6 days followed by rare mosaic expression in the anterior CNS (henceforth referred to as p53EPS, **Fig 1A** [↗](#)), as expected from a transient mosaic injection strategy used to express stable concatemers of DNA vectors (Langenau et al., 2008 [↗](#)). At 15 days post fertilization (dpf), this transgene combination led to visible mScarlet-positive brain lesions in the anterior CNS of live zebrafish (**Fig 1B-D** [↗](#)). Injections of single linearized vector and vector combinations resulted in a maximum incidence of approximately 15-20% affected zebrafish induced by the p53EPS combination by 45 dpf (**Fig 1D** [↗](#)). Inter-tumoral variability was observed among p53EPS mosaic-

injected zebrafish, with tumors of variable size arising in different brain regions including the telencephalon and diencephalon regions (8/29 = 27.5%, from three independent tumor screens), and the optic tectum/mesencephalon region (21/29 = 72.5%, from three independent tumor screens) (**Supplementary Fig 1** [↗](#)). To further define the tissue of origin of zebrafish p53EPS CNS lesions, we co-injected linearized *gfap*:EGFP ([Don et al., 2017](#) [↗](#)) and visualized EGFP expression in 100% of brain masses at 30–40 dpf (**Fig 1A,B** [↗](#)), supporting glial identity and a novel *in vivo* model of malignant glioma.

To test for transformation of p53EPS cells from primary mosaic-injected animals, we harvested bulk tissue from dissected zebrafish brains and transplanted dissociated cells into the hindbrain ventricles of 2 dpf syngeneic *tp53* wild-type (CG1) zebrafish ([Mizgireuv and Revskoy, 2006](#) [↗](#)). At approximately 18 days post-transplant (18 dpt, or 20 dpf), we screened for mScarlet+ fluorescence and visualized *her4.1*/gfap+ tumor cell outgrowth in 16–25% of transplanted hosts (n=3 independent screens, **Fig 1E,F** [↗](#)), supporting oncogenic transformation and malignant growth *in vivo* of p53EPS cells. Fluorescence activated cell sorting (FACS) revealed fluorescently labeled tumor cells in un-labeled syngeneic host brains and co-expression of *her4.1*:mScarlet/*gfap*:EGFP (**Fig 1G** [↗](#)), further supporting tumor growth from a glial-derived progenitor cell, which was expected from *her4.1*-specific oncogene activation during zebrafish development ([Than-Trong et al., 2020](#) [↗](#)).

We performed serial sectioning of uninjected control and primary p53EPS zebrafish brains and performed histological characterizations, which revealed highly aggressive and proliferative tumors in p53EPS consisting of a heterogeneous mix of different malignant cell types, compared to normal brains (**Fig 1H–O** [↗](#), **Supplementary Fig 2** [↗](#)). One major cell type displayed a round granular-like morphology with a high nuclear to cytoplasm ratio, while a second cell type found within the same tumor masses displayed a flattened and elongated morphology (**Fig 1L** [↗](#), **Supplementary Fig 2** [↗](#)). In general, tumors displayed clean margins without invasive boundaries, necrosis, or micro vascularization; however, a few instances of cell migration along vasculature were detected. Certain p53EPS tumors displayed embryonal features including rosette-type structures and a high mitotic index, overlapping histological features with glioblastoma with primitive neuronal component ([Suwala et al., 2021](#) [↗](#)), which may be expected given our chosen combination of molecular drivers as well as a neural progenitor and/or radial glial cell of origin, during developmental stages.

To assess oncogenic signaling pathway activation, we stained unaffected and p53EPS tumor-positive brain sections for phosphorylated-ERK (p-ERK) and phosphorylated-AKT (p-AKT), indicators of activated MAPK and AKT signaling pathways, respectively (**Fig 1J–K, N–O** [↗](#)). Compared to normal brains, we found increased p-ERK and p-AKT staining within tumor lesions, consistent with elevated MAPK and AKT activation driving malignant transformation (**Fig 1N–O** [↗](#)). To further validate effects of relevant downstream signaling pathway activation, we also co-injected linearized *her4.1*:KRAS^{G12D} + *her4.1*:PI3KCA^{H1047R} + *her4.1*:mScarlet into *tp53*^{−/−} embryos (p53KPS). At 15–20 dpf, we visualized tumor onset and penetrance comparable to p53EPS (**Supplementary Fig 3** [↗](#)), suggesting a dominant role for MAPK/AKT pathway activation downstream of RTK signaling in driving tumor formation, and a flexible oncogenic strategy for inducing robust brain tumor formation in zebrafish.

Gene set enrichment analyses (GSEA) reveal gene expression patterns consistent with human mesenchymal glioblastoma and inflammation

To further characterize our novel zebrafish brain tumor models, we performed bulk RNA sequencing (RNAseq) of three independent p53EPS tumor-burdened brains, three independent p53KPS tumor-burdened brains, and three age-matched, control-injected zebrafish brains that remained tumor-free at the time of harvesting tissue. Using hierarchical clustering on normalized

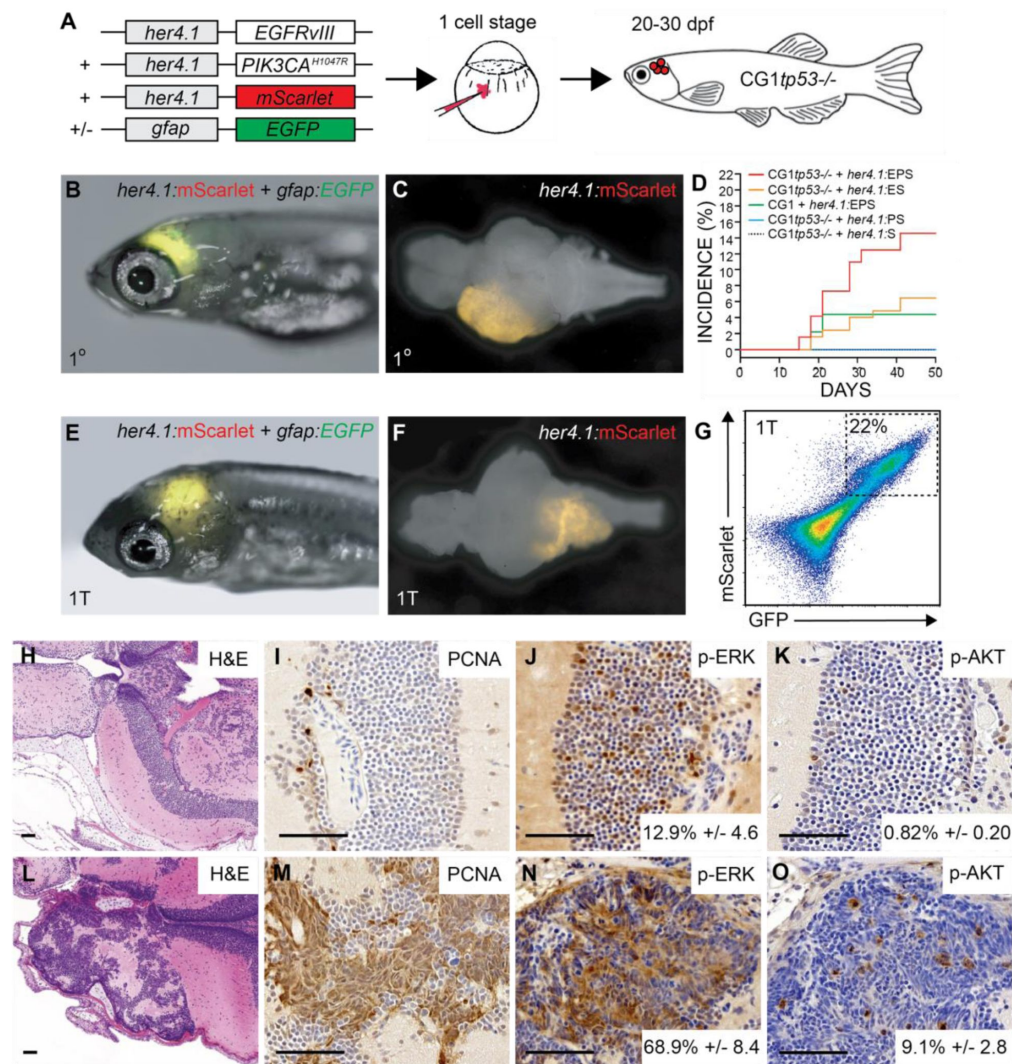


Figure 1.

***her4.1*-driven over-expression of patient-relevant oncogenes drives glial-derived brain tumor formation in syngeneic *tp53* loss-of-function mutant zebrafish.**

(A) Schematic of modeling strategy where linearized transgene vectors with the zebrafish *her4.1* promoter driving human EGFR^{vIII}, human PI3KCA^{H1047R}, and mScarlet fluorescent protein are co-injected at the one-cell stage into syngeneic (CG1 strain) *tp53*^{-/-} mutant zebrafish embryos. Starting at 15 days post fertilization (dpf), mosaic-injected zebrafish were screened for CNS tumor formation, indicated by mScarlet expression in the brain region of live zebrafish. Co-injection of *gfap*:GFP linearized transgene is used to assess glial-specific cell fate specification *in vivo*. (B) *her4.1*:mScarlet and *gfap*:GFP expression in the anterior CNS of mosaic-injected syngeneic (CG1 strain) *tp53*^{-/-} zebrafish at 30dpf. (C) Whole brain dissected from a p53EPS mosaic-injected zebrafish at 30dpf. (D) Cumulative frequencies of mScarlet+ CNS lesions in syngeneic *tp53*^{-/-} mutant (CG1tp53^{-/-}) and wild-type (CG1) zebrafish injected at the one cell stage with *her4.1*:EGFR^{vIII} (E), *her4.1*:PI3KCA^{H1047R} (P), and/or *her4.1*:mScarlet (S). (E) Syngeneic (CG1 strain) zebrafish at 30dpf engrafted with *her4.1*:mScarlet+/*gfap*:GFP+ brain tumor cells, following primary transplantation (1T) at 2dpf into the embryonic brain ventricle. (F) Whole brain dissected from engrafted syngeneic host (CG1) zebrafish at 30dpf. (G) FACS plot of bulk syngeneic host brain following primary transplant (1T) of *her4.1*:EGFR^{vIII} + *her4.1*:PI3KCA^{H1047R} + *her4.1*:mScarlet + *gfap*:GFP brain tumor cells. (H-O) Histological staining of uninjected control (H-K) and p53EPS tumor burdened brains (L-O). (H,L) Hematoxylin and eosin (H&E) staining of coronal sections highlighting telencephalon and diencephalon regions of representative control (H) and p53EPS (L) brains. (I,M) Proliferating Cell Nuclear Antigen (PCNA) staining of control (I) and p53EPS (M) brain sections. (J,N) Phosphorylated-ERK (p-ERK) staining and quantifications reveal increased MAPK signaling pathway activation in p53EPS tumors ($p < 0.001$, $n = 3$ independent tumor sections). (K,O) Phosphorylated-Akt (p-Akt) staining and quantifications reveal increased Akt signaling pathway activity in p53EPS tumors ($p = 0.007$, $n = 3$ independent tumor sections). Scale bars represent 50µm.

gene expression, our tumor-free control samples clustered together and apart from p53EPS and p53KPS tumor brains, which displayed a recognizable level of transcriptional variability across principal components (**Fig 2A**). Interestingly, p53EPS and p53KPS samples failed to cluster according to oncogenic drivers (**Fig 2A**), suggesting molecular similarities as well as inter-tumor heterogeneity reflecting differences in tumor location, size, and contribution of tumor cells to total sample inputs.

Given the relevance of our p53EPS driver combination to human glioblastoma, we chose to focus the remainder of our molecular analyses on p53EPS samples. Using differential gene expression analysis, we identified a conserved set of differentially expressed (DE) genes in p53EPS tumor brains, with 236 significantly upregulated and 28 downregulated genes, compared to control injected brains at 20-30 dpf ($2 > \log_2\text{foldChange} > -2$, adjusted p -value < 0.05 , Supplementary Table 1, Supplementary Table 2, **Fig 2B**). Using human orthologs of differentially expressed genes, we performed Gene Set Enrichment Analysis (GSEA) (Mootha et al., 2003; Subramanian et al., 2005), comparing our p53EPS zebrafish model to published expression patterns for human glioblastoma subtypes, as well as embryonal brain tumors including designated subtypes of medulloblastoma (Cavalli et al., 2017; McLendon et al., 2008; Wang et al., 2017). Among these gene sets, we found a significant enrichment for the mesenchymal subtype of human glioblastoma in p53EPS differentially expressed genes (**Fig 2C**, Supplementary Table 3) (McLendon et al., 2008; Wang et al., 2017). In contrast, while certain genes for other glioblastoma and/or medulloblastoma subtypes were upregulated in p53EPS samples, no significant enrichment was found for these specific gene expression signatures (**Supplementary Fig 4**, Supplementary Table 3) (Cavalli et al., 2017; McLendon et al., 2008; Wang et al., 2017).

To assess potential underlying molecular mechanisms involved in p53EPS formation, we assessed enrichment for Hallmark gene expression sets available through the Molecular Signature Database (MSigDB, Supplementary Table 4) (Liberzon et al., 2011). Interestingly, 7 of the top 13 enriched gene sets identified (NOM p -value < 0.05) related to inflammation or inflammatory signaling pathways, including the Interferon gamma response, TNFA signaling, the Interferon alpha response, and Jak/STAT3 signaling (**Fig 2D**, Supplementary Table 4), suggesting a strong inflammatory component in our brain tumor model. Additional pathway signatures included those related to RAS signaling, Hypoxia, and Epithelial-to-Mesenchymal Transitions (EMT) (Supplementary Table 4), suggesting on target oncogenic pathway activation, hypoxia, and invasive properties, consistent with aggressive glioblastoma (Majc et al., 2020; Park and Lee, 2022). Therefore, bulk RNA sequencing expression data supports significant molecular similarity between our zebrafish p53EPS brain tumor model and human glioblastoma of the mesenchymal subtype, with a significant inflammatory component.

Given the *in vivo* context and contribution of tumor and non-tumor cell types to our bulk RNAseq analysis, we decided to assess RNA expression in FACS-sorted *her4.1*:mScarlet-positive tumor cells, as well as *her4.1*:mScarlet-negative bulk stromal cells, compared to non-tumor whole brain tissue (**Fig 2E**, Supplementary Table 5). Interestingly, we observed increased immune cell and inflammatory gene expression in both p53EPS tumor and stromal cell fractions compared to control unaffected whole brain tissue, including transcripts associated with myeloid cell types (*mpeg1*, *irf7*, *irf8*), lymphoid progenitor and specific subtype regulators (*rag1*, *rag2*, *lck*), *stat1a/b*, *fas* cell surface death receptor, and toll-like receptor 4b (*tlr4bb*), among others genes involved in immune responses (adjusted p -value < 0.0001 , **Fig 2E**, Supplementary Table 5). We validated a selection of genes from RNAseq using RT-PCR (**Supplementary Fig 5**), as well as transient co-injection of linearized transgene with our EPS mix to assess tumor cell-specific expression of lymphoid *rag2*:EGFP and myeloid *mpeg1.1*:EGFP at 30 dpf (**Fig 2F-I**). Interestingly, we observed co-expression of *her4.1*:mScarlet/*rag2*:EGFP and *her4.1*:mScarlet/*mpeg1.1*:EGFP in p53EPS tumors (**Fig 2F-I**), suggesting promoter activity at certain immune-associated genes within established p53EPS tumor cells *in vivo*. These data suggest inflammation-associated gene expression in both

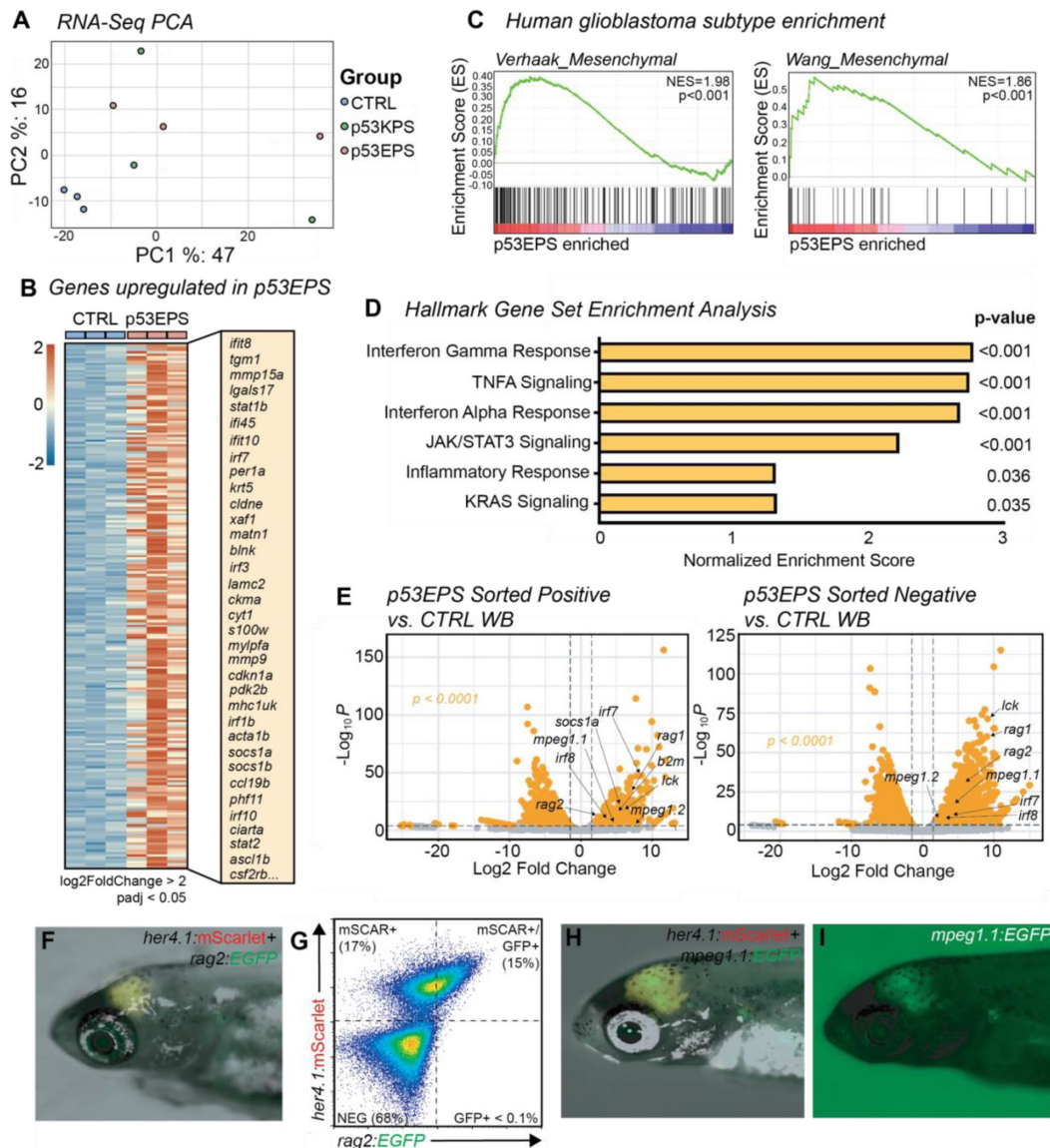


Figure 2.

RNA expression analysis establishes enrichment of mesenchymal glioblastoma and inflammation signatures in p53EPS model.

(A) Principal component analysis (PCA) of mRNA sequencing from whole control injected brains (CTRL), p53EPS and p53KPS tumor-burdened brains. (B) Heatmap of normalized counts for genes upregulated in p53EPS tumor-burdened brains (log2FoldChange > 2, padj < 0.05), compared to whole control injected brains (CTRL). A selected list of upregulated transcripts is indicated. (C) Gene set enrichment analysis (GSEA) plots of published gene signatures for mesenchymal subtype glioblastoma for genes differentially regulated in p53EPS compared to control injected brains (McLendon et al., 2008; Wang et al., 2017). Normalized Enrichment Scores (NES) and nominal p-values are indicated. (D) Bar plot of Normalized Enrichment Scores from GSEA of Hallmark gene sets (Liberzon et al., 2011). (E) Volcano plots of differentially expressed genes between sorted mScarlet+ p53EPS tumor cells and control injected whole brain tissue (CTRL WB), as well as between sorted mScarlet-negative cells from p53EPS tumor-burdened brains and control injected whole brains (CTRL WB). (F) *her4.1:mScarlet* and *rag2:EGFP* expression in live zebrafish with a p53EPS tumor at 30 dpf. (G) FACS plot of p53EPS brain with *rag2:EGFP* co-expression from (F). (H, I) *her4.1:mScarlet* and *mpeg1.1:EGFP* expression in live zebrafish with a p53EPS tumor at 30 dpf.

zebrafish glioblastoma-like cells and the TME and is consistent with inflammatory gene expression in tumor cells associated with immune evasion and *in vivo* growth in other models (Gangoso et al., 2021; Yang et al., 2022). Altogether, our *in vivo* expression data supports p53EPS zebrafish as a comparable and relevant model system to study glioblastoma tumor biology, as well as intercellular interactions within an endogenous TME.

Zebrafish mesenchymal glioblastoma-like tumors recruit activated mononuclear phagocytes at early stages of tumor formation

Given that p53EPS tumor-burdened brains displayed strong enrichment for transcriptional signatures associated with inflammatory signaling, we were interested in the role for innate immune cells in the earliest stages of p53EPS tumor formation. Myeloid cells including microglia and macrophages are recruited in many different subtypes of primary brain tumors and in brain metastases (Gutmann and Kettenmann, 2019; Khan et al., 2023). However, how phagocyte populations affect glioblastoma initiation is less well understood given that most studies utilize established tumor cell models and/or patient derived tissue xenograft transplantations into immune-compromised host animals. To first determine whether myeloid-derived phagocytic cell lineages are enriched in p53EPS lesions at early stages of tumor initiation, we used neutral red staining that labels lysosomal-rich phagocytes, irrespective of cell type (Fig 3A-C) (Herbomel et al., 2001; Shiao et al., 2015). At 10 dpf, we observed enrichment of neutral red-positive foci in regions of *her4.1*:mScarlet+ fluorescent intensity (Fig 3B,C), suggesting phagocyte infiltration during p53EPS initiation, prior to typical observation of macroscopic tumor masses (Fig 1D). To assess the activation state of phagocytes in and surrounding p53EPS tumors, we co-injected our oncogene combination into *tp53*^{-/-} embryos carrying a Tg(*tnfa*:EGFP) transgenic marker of activated and pro-inflammatory phagocytes (Hao et al., 2012; Nguyen-Chi et al., 2015), and observed enrichment of *tnfa*:EGFP single-positive cells in *her4.1*:mScarlet+ lesions (Fig 3D,E). We also performed GSEA against immunologic signature gene sets from the Molecular Signatures Database (MSigDB, (Liberzon et al., 2011; Mootha et al., 2003; Subramanian et al., 2005)), and identified multiple enriched gene expression signatures associated with inflammation and inflammatory cell types in p53EPS tumor-burdened brains (Supplementary Table 6). p53EPS displayed features of both pro- and anti-inflammatory gene expression, which is consistent with a mixed immune activation/suppressive state in glioblastoma known to play an important role in tumor progression in patients (Biswas et al., 2008; Karimi et al., 2023; Quail and Joyce, 2017; Ren et al., 2023). However, p53EPS tissues more significantly enriched for gene expression associated with activated and pro-inflammatory phagocytes including classical M1 macrophages, compared to alternative M2 polarized macrophages, and genes typically downregulated during M2 polarization following macrophage colony stimulating factor (M-CSF) treatment (Supplementary Fig 6, Supplementary Table 6), further suggesting strong promotion of inflammation within the TME.

Given enrichment for *tnfa*:EGFP+ phagocytes and signatures associated with pro-inflammatory macrophages in our glioblastoma model, we decided to investigate microglia/macrophages cell dynamics *in vivo* using live confocal imaging following co-injection of the linearized transgene combination into *tp53*^{-/-}; Tg(*mpeg1.1*:EGFP) zebrafish (Ellett et al., 2011). We first looked closely at a time-course of lesion formation at 5, 7, and 9 dpf relative to microglia in the zebrafish brain (Fig 3F-K), which colonize the zebrafish neural retina by 48 hpf, and the optic tectum by 84 hpf (Herbomel et al., 2001). At 5 dpf (120 hpf), we observed comparable levels of *mpeg1.1*:EGFP+ microglia throughout the cephalic region and in the brain in both control and p53EPS injected zebrafish (Fig 3F,I). p53EPS-injected zebrafish brains displayed mosaic cellular expression of *her4.1*:mScarlet; however, cells were sparse and diffuse, and no large intensely fluorescent tumor-like clusters were detected at this stage (Fig 3I, n=12/12 tumor negative). At 7 dpf, we detected small clusters of *her4.1*:mScarlet+ cells (Fig 3J, n=5/9), indicative of early lesion formation. Interestingly, these lesions were surrounded and/or infiltrated by *mpeg1.1*:EGFP+ microglia, while

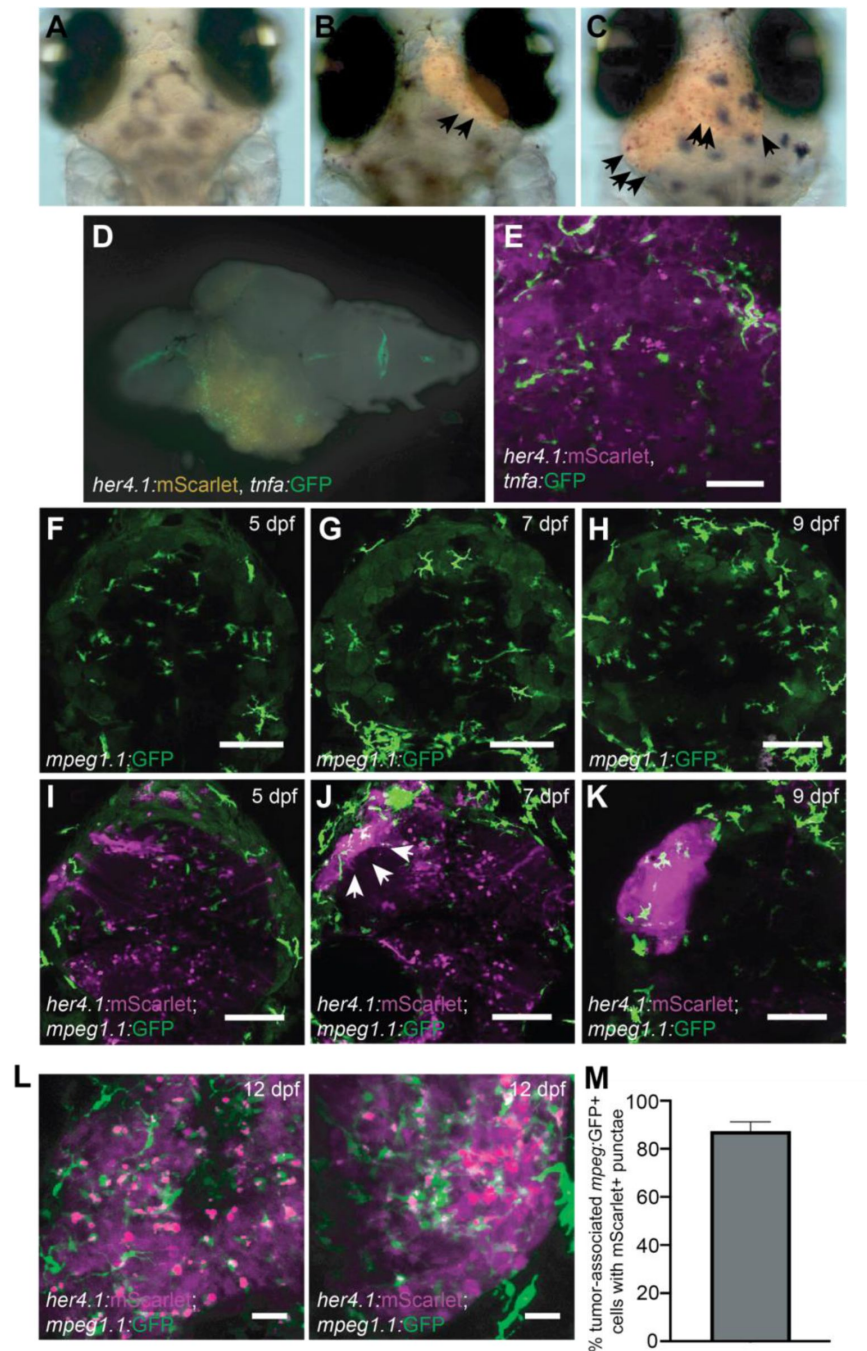


Figure 3.

p53EPS recruits activated microglia/macrophages at early stages of tumor initiation.

(A-C) Neutral red staining of p53EPS mScarlet tumor-negative (A) and mScarlet tumor-positive brains (B,C) at 10 dpf. Neutral red foci in early-stage lesions are highlighted with arrows and are indicative of phagocytic cells. (D) Whole brain with p53EPS-induced tumor in a transgenic *Tg(tnfa:EGFP)* zebrafish at 20 dpf. (E) Z-stack projection of live confocal imaging of p53EPS tumor in transgenic *Tg(tnfa:EGFP)* background. (F-H) Z-stack projections of control uninjected (F-H) and p53EPS brains (I-K) at 5 dpf (F,I), 7 dpf (G,J), and 9 dpf (H,K) in transgenic *Tg(mpeg1.1:EGFP)* background. White arrows highlight an early-stage p53EPS lesion and associated *mpeg1.1:EGFP*⁺ cells. (L) Z-stack projections of two independent p53EPS brains at 12 dpf in transgenic *Tg(mpeg1.1:EGFP)* background. (M) Quantification of tumor-associated *mpeg1.1:EGFP*⁺ cells with overlapping and/or internalized *her4.1:mScarlet*⁺ punctae (n=3 independent tumors).

at 9 dpf *her4.1*:mScarlet+ expression became highly specific to tumor lesions, and these lesions were consistently associated with *mpeg1.1*:EGFP+ microglia/macrophages (Fig 3K, n=8/8 tumor-positive zebrafish).

At 12 dpf in p53EPS zebrafish with established tumors, we observed fluorescent *mpeg1.1*:EGFP+ cells surrounding and within regions of concentrated *her4.1*:mScarlet+ fluorescence (Fig 3L, Movie 1). Microglia/macrophages outside of early-stage *her4.1*:mScarlet+ lesions displayed highly ramified morphologies, with several processes that were extended and retracted, indicative of environmental surveillance (Fig 3L, Movie 1) (Nimmerjahn A et al., 2005). *mpeg1.1*:EGFP+ microglia/macrophages infiltrated into dense *her4.1*:mScarlet+ regions displayed more rounded and amoeboid-like morphology, supporting their activation in association with p53EPS oncogenic cells (Karperien et al., 2013). Interestingly, *mpeg1.1*:EGFP+ microglia/macrophages dynamically interacted with *her4.1*:mScarlet+ cells (Movies 1 and 2), and in p53EPS oncogenic masses, *mpeg1.1*:EGFP+ microglia/macrophages associated closely with and often displayed internalized *her4.1*:mScarlet+ punctate cells (Fig 3L,M), suggesting engulfment and removal of p53EPS cells during tumor formation *in vivo*. Together with our expression data, visualization of infiltrating myeloid-derived immune cells in p53EPS glioblastoma-like lesions, dynamic *mpeg1.1*:EGFP+ microglia/macrophages-p53EPS interactions, and tumor cell engulfment suggests anti-tumoral activity at early stages that could negatively affect tumor formation *in vivo*.

Inflammation-associated *irf7* and *irf8* are required to inhibit p53EPS tumor formation *in vivo*

Interferon regulatory factor (Irf) proteins regulate transcription of interferon genes and support a variety of different immune responses. Irf7 and Irf8 are critical for global activation of the type I IFN response following stimulation and for myeloid cell development, respectively (Günthner and Anders, 2013; Ning et al., 2011; Shiau et al., 2015). Irf8 is a conserved determinant of macrophage cell fate, as well as dendritic cell survival and function, and *irf8* mutant zebrafish lack microglia/macrophages in the brain up to 31 dpf (Shiau et al., 2015; Sichien et al., 2016). Irf7 also drives differentiation of macrophages; however, in zebrafish and other systems, Irf7 is more broadly activated in immune cells in response to infection, with evidence also supporting non-immune cell-related functions during development and cancer progression (Feng et al., 2016; Gangoso et al., 2021; Günthner and Anders, 2013; Hu et al., 2022; Ning et al., 2011; Yang et al., 2022). Interestingly, we observed upregulation of several interferon regulatory factor (Irf) family members including *irf7* and *irf8* in p53EPS tumor-burdened brains (Fig 2B, Supplementary Table 1, 2), suggesting potential roles for these master regulators of inflammation on immune-related responses during p53EPS tumorigenesis *in vivo*. Therefore, to assess functional roles for Irf7 and Irf8 in p53EPS formation, we used a transient CRISPR/Cas9 gene targeting approach to knock-down *irf7* or *irf8* genes prior to p53EPS tumor formation using co-injection of 2-3 gRNAs each targeting *irf7* or *irf8* (Supplementary Table 7), together with Cas9 protein and linearized EPS into one-cell stage *tp53*^{-/-} embryos. At 2-5 dpf, we extracted DNA from a subset of injected embryos and observed a gene targeting efficiency of >90% and >65% INDELS at the *irf7* or *irf8* loci, respectively (ICE analysis, ice.synthego.com, Synthego, 2019). Gene transcript knock-down was also verified using RT-PCR (Supplementary Fig 7). Consistent with previous reports, *irf8* knock-down resulted in significant reductions in neutral red-positive phagocytes at 8-10 dpf (Supplementary Fig 8) (Shiau et al., 2015), and while no significant differences in neutral red-positive phagocyte number were observed following *irf7* gene targeting, *irf7* CRISPR/Cas9-injected animals displayed early mortality unrelated to brain tumor formation, with ~80% of CRISPR-injected animals displaying illness prior to two months of age, consistent with a broad requirement for Irf7 in immune responses, among other functions.

Remarkably, following *irf7* or *irf8* gene knock-down with p53EPS, we observed robust tumor formation, with 65% of *irf7*-targeted and 42% of *irf8*-targeted p53EPS zebrafish developing tumors by 30 dpf, compared to 20% p53EPS incidence (Fig 4A-D, *irf7* $p < 0.0001$, *irf8* $p = 0.0155$, Fisher's

exact test), suggesting an important inhibitory role for *irf7* and *irf8* in p53EPS tumor initiation. In p53EPS tumor brains with *irf7* and *irf8* knock-down, we observed reduced expression of genes associated with our inflammation signature in p53EPS tumor-burdened brains, including immune evasion-associated transcripts like *suppressor of cytokine signaling 1a (socs1a)* (**Supplementary Fig 7**), suggesting reduced tumor-specific inflammation and associated immune-evasion mechanisms, which were previously shown to be upregulated in response to anti-tumor cell infiltration (Gangoso et al., 2021). To further validate gene targeting and assess the spatial localization of *irf7* expression within the TME, we performed fluorescent *in situ* hybridization (FISH) in control and *irf7* knock-down tumors and detected robust transcript expression within early-stage p53EPS lesions at 8 dpf (**Fig 4E**). In animals co-injected with *irf7* CRISPR, we detected p53EPS tumor formation, albeit with reduced *irf7* in and surrounding tumor cells (**Fig 4E**), suggesting potential roles for *irf7* in inhibiting p53EPS initiation in tumor cells and/or within the TME.

Given broad *Irf* gene expression in p53EPS tumors (**Fig 2E**, **Supplementary Fig 5, 7, 9**, **Supplementary Table 5**), as well as reports of *IRF7* gene expression in human glioblastoma tumor cells, association with worsened patient outcome, and potential roles in tumor stem cell biology (Jin et al., 2012; Yang et al., 2022), we decided to investigate tumor cell-specific roles for *Irf* genes in p53EPS initiation. To specifically knock-down *irf7* in p53EPS tumor cells, we generated a transgenic *Tg(her4.1:Cas9-2A-EGFP);tp53^{-/-}* zebrafish strain and co-injected embryos with linearized EPS + gRNAs targeting *irf7* (**Fig 4F**). Despite a gene targeting efficiency of ~28% in pooled tumor cells from 5 tumor-burdened animals, we did not observe any significant changes in p53EPS tumor formation in Cas9-2A-EGFP-positive zebrafish compared to Cas9-2A-EGFP-negative control injected siblings (**Fig 4F-H**). We observed similar effects following co-injection of gRNAs targeting *irf8* (**Fig 4G**), suggesting that increased p53EPS tumor initiation following *Irf* gene knock-down is a consequence of *irf7* and *irf8* loss-of-function in the TME.

Phagocyte activity suppresses p53EPS engraftment following transplant into syngeneic host zebrafish

Given conserved roles for *irf8* across species in the specification and function of different phagocyte populations (Shiau et al., 2015), our functional data suggested an anti-tumor role for *irf*/phagocyte activity in the TME during p53EPS tumor initiation. To assess the role more specifically for phagocyte populations within the TME, we decided to transplant p53EPS tumor cells into the hindbrain ventricle of syngeneic host embryos at 2 dpf. Importantly, serial transplantation of tumor cells is commonly used as an experimental surrogate to assess relapse potential (Blackburn et al., 2011; Hayes et al., 2018; Ignatius et al., 2017; Smith et al., 2010), allowing us to use p53EPS engraftment and re-growth to study the tumor propagating properties of p53EPS within different TMEs. To test the role for phagocytes in engraftment, we transplanted dissociated bulk tumor cells into syngeneic embryos along with Clodronate (Clodronate liposomes, Clodrosomes), a chemical used to eliminate phagocytes *in vivo*, and microglia following intracerebral injection (Andreou et al., 2017; Fernández-López et al., 2016; Hanlon et al., 2019; Yan et al., 2019; Yang et al., 2021). At 18 days post-transplant (20 dpf), we observed 50% p53EPS tumor cell engraftment in host animals co-injected with Clodronate, compared to 23% injected with vehicle control liposomes ($p=0.0048$, Fisher's exact test, **Fig 5A-C**). Interestingly, while p53EPS engrafted brains maintained inflammatory gene expression patterns like primary p53EPS tumor-burdened brains (**Supplementary Fig 9A**), co-transplantation with Clodronate inhibited inflammatory gene expression in bulk tissue (**Supplementary Fig 9B**, normalized to mScarlet expression to control for differences in tumor size), suggesting that reduced phagocyte-driven inflammation supports p53EPS tumor cell engraftment and growth. We also performed p53EPS bulk tumor cell transplantations into the hindbrain ventricles of *irf8* CRISPR/Cas9-injected syngeneic host zebrafish embryos at 2 dpf and observed 46% p53EPS tumor cell engraftment at 20 dpf, compared to 19% engraftment in *irf8* wild-type host zebrafish ($p=0.0002$, Fisher's exact test, **Fig 5D**), further supporting an inhibitory role

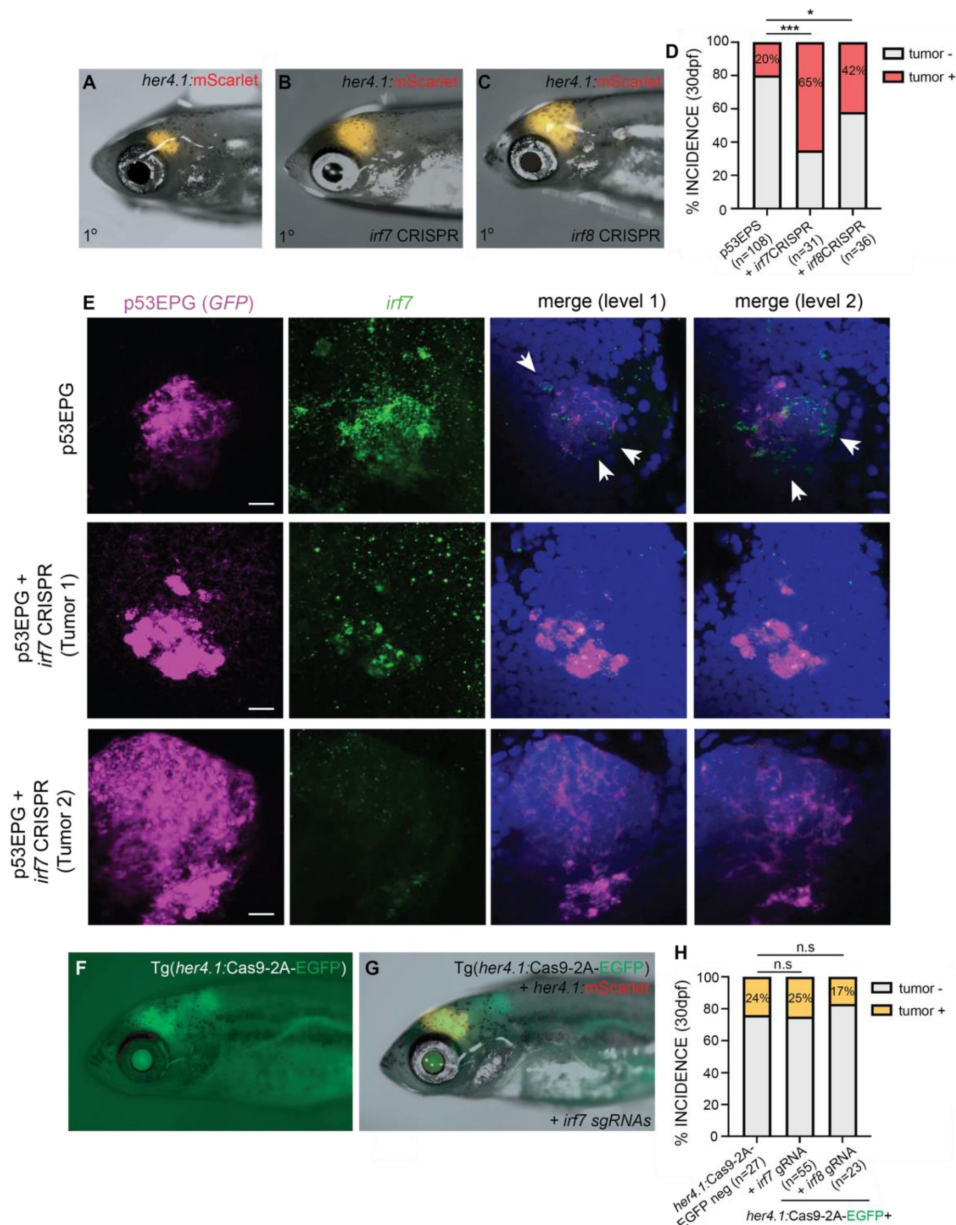


Figure 4.

Inflammation-associated *irf7* and *irf8* inhibit p53EPS formation *in vivo*.

(A–C) Primary (1°) control (A), *irf7* CRISPR/Cas9 (B), and *irf8* CRISPR/Cas9 (C) injected p53EPS at 30 dpf. (D) p53EPS incidence at 30 dpf in control (n=3 independent experiments, 108 zebrafish), *irf7* CRISPR/Cas9 (***p<0.0001, Fisher's exact test, n=2 independent experiments, 31 zebrafish), and *irf8* CRISPR/Cas9 (*p=0.0155, Fisher's exact test, n=2 independent experiments, 36 total injected zebrafish). (E) Representative fluorescent in situ hybridization (FISH) images of whole mount control p53EPG (EGFR^{VIII} + PI3KCA^{H1047} + GFP) and p53EPG + *irf7* CRISPR/Cas9-injected zebrafish at 8 dpf. p53EPG (GFP, magenta) and *irf7* (green) images represent Z-stack projections through tumor lesions (11 optical sections each). Merged images represent single optical sections at two spatially separated levels within control and *irf7* knock-down tumors. DAPI staining (blue) is used to label nuclei. White arrowheads highlight *irf7* expression specific to the TME. Scale bars represent 10μm. (F) Tg(*her4.1:Cas9-2A-EGFP*) expression at 30 dpf. (G) mScarlet+ p53EPS at 30 dpf in Tg(*her4.1:Cas9-2A-EGFP*) injected with *irf7* gRNAs at the one-cell stage. (H) p53EPS incidence at 30 dpf in Tg(*her4.1:Cas9-2A-EGFP*)-negative gRNA-injected control siblings, and Tg(*her4.1:Cas9-2A-EGFP*) zebrafish injected at the one-cell stage with *irf7* or *irf8* gRNAs. n.s. not significant, Fisher's exact test.

for phagocytes within the TME. Altogether, our modeling data using patient-relevant oncogene combination in *tp53* loss-of-function background demonstrates an important role for inflammation in glioblastoma initiation and relapse including inhibitory roles for Irf-dependent signaling pathways, which in part may be attributed to anti-tumoral phagocyte activity within the TME.

Discussion

Genetically engineered mouse models (GEMM) are widely used for the study of tumorigenesis in a physiological context; however, most cannot fully recapitulate the genetic heterogeneity found in glioblastoma in a timely and cost-effective manner and are therefore limited in the context of preclinical drug testing. Patient-derived xenografts (PDX) resemble patient tumors more closely as they retain mutational heterogeneity; however, PDX models cannot be used to address mechanisms of tumor onset in an intact and endogenous tumor microenvironment given that they are derived from pre-evolved tumor tissue and are either studied *ex vivo* or engrafted into immune-deficient animal hosts. Human stem cell-derived organoids represent a great advance in the field for the study of tumor development and for screening new therapeutic strategies, but the lack of an intact TME excludes both known and unknown inter-cellular interactions, which are increasingly appreciated to influence tumor cell evolution and drug responses in patients. Therefore, to address current limitations with respect to modeling human glioblastoma *in vivo*, we used zebrafish and neural stem/progenitor-specific expression of common oncogenic variants in a *tp53*-deficient background. We generated a robust tumor model, with a gene expression signature consistent with human glioblastoma of the mesenchymal subtype. We also harnessed the unique advantages of zebrafish modeling including *in vivo* live imaging and rapid genetic manipulations to assess characteristic inflammatory gene expression signatures within the TME, as well as roles for certain inflammatory cell types at tumor initiation and relapse following transplantation into syngeneic host animals.

Several animal models including zebrafish models of human brain cancer have been generated using transgenic overexpression of human oncogenes or xenograft transplantation of patient-derived tumor cells into immune-compromised hosts (Almstedt et al., 2022 [4](#); Ju et al., 2015 [5](#); Mayrhofer et al., 2017 [6](#); Modzelewska et al., 2016 [7](#)), and these models have proven useful for defining important molecular pathways in glioma transformation, tumor growth, and cell migration. However, most of these available zebrafish models are induced using single oncogenic drivers driven by a variety of neuronal and/or glial-specific promoters at different time points throughout development, with the use of single oncogenes in a functional *tp53* signaling background potentially contributing to their inability to model highly aggressive and relapse-like phenotypes *in vivo*. Xenograft transplantations of human glioblastoma cells into zebrafish larvae omits a fully functional immune system, which our work suggests plays an important role on tumor initiation and growth both in the primary context and following p53EPS tumor cell transplantation. Therefore, our zebrafish model of aggressive and transplantable human glioblastoma represents an important opportunity to study aspects related to tumor cell progression and relapse in an intact TME.

One important consideration with model generation is use of genetic drivers that are reflective of variants found in patient tumors. EGFR^{VIII} is a common genetic variant in glioblastoma that we show contributes to aggressive brain tumor formation in our zebrafish model. Interestingly, we found that our p53EPS model significantly enriches for gene expression signatures associated with mesenchymal glioblastoma, which may be unexpected given our use of EGFR^{VIII} as a driver (Verhaak et al., 2010 [8](#)). However, from our p53EPS transcriptional signatures, we also detected expression of several markers associated with EGFR^{VIII} and the classical subtype including neural progenitor and neural stem cell genes (**Supplementary Fig 4** [4](#), [5](#); **Supplementary Table 1-5**), potentially suggesting underlying molecular heterogeneity. In addition, histological

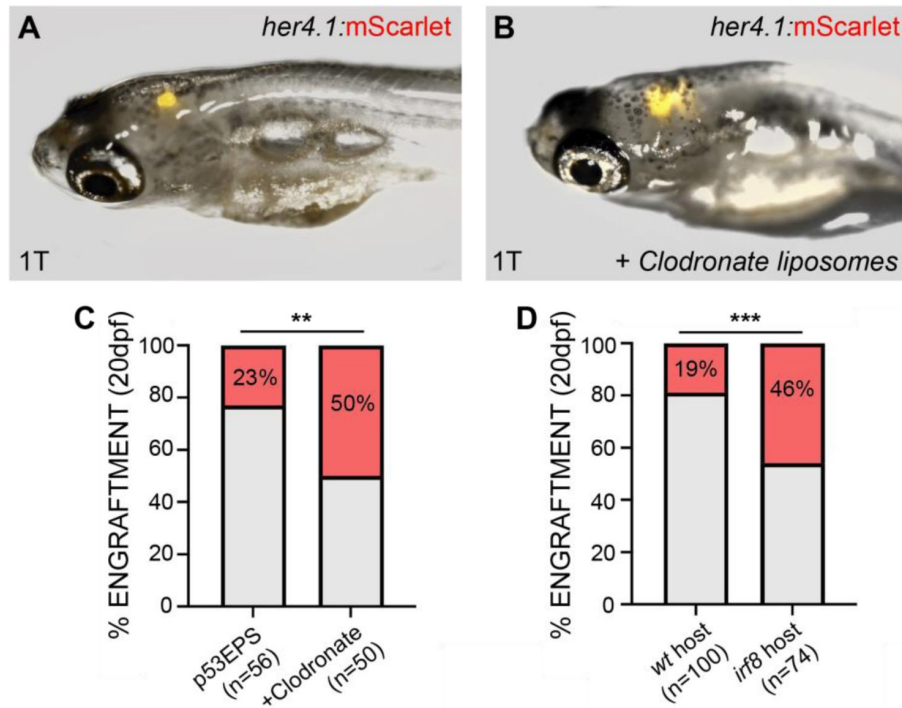


Figure 5.

Inflammation-associated phagocytes inhibit p53EPS tumor engraftment.

(A,B) CG1 syngeneic host zebrafish at 20 dpf engrafted with p53EPS tumor cells transplanted with vehicle control (A) or Clodronate liposomes (B) at 2 dpf. **(C-E)** Quantification of p53EPS control engrafted and p53EPS tumors engrafted into CG1 host embryos with (C) Clodronate liposomes ($p=0.0048$, Fisher's exact test, $n=2$ independent experiments, total 56 transplanted vehicle control and 50 transplanted Clodrosome-injected hosts), (D) engrafted into *irf8* CRISPR/Cas9-injected into CG1 syngeneic host embryos ($p=0.0002$, Fisher's exact test, $n=2$ independent experiments, total 100 transplanted control and 74 transplanted *irf8* CRISPR/Cas9-injected hosts).

characterization of p53EPS tumors reveal several embryonal-type features, suggesting pathological differences with respect to over-expressing human oncogenes during zebrafish development, which likely affect certain morphological features and/or gene expression patterns compared to those seen in adult glioblastoma patients. A significant amount of heterogeneity is seen in patients with respect to gene expression and histopathology, and our characterizations are consistent with certain zebrafish and rodent models using MAPK/AKT pathway activation within the developing nervous system (Huse and Holland, 2009 [↗](#); Jin et al., 2021 [↗](#); Mayrhofer et al., 2017 [↗](#); Wei et al., 2006 [↗](#)). However, in the future it will be interesting to use our model and assess temporal and driver-specific contributions to specific pathohistological features and molecular subtypes at early and late stages of tumor progression, as well as inter-tumoral heterogeneity and subtype transitions to better understand how phenotypic plasticity contributes to glioblastoma growth and relapse in our model, and in patients.

Zebrafish offer a *in vivo* platform for live imaging tumor cell dynamics at high resolution, and we demonstrated a powerful model to study inter-cellular interactions during glioblastoma initiation within an endogenous TME, an approach that is not currently feasible in rodent models. The contributions of macrophages/microglia and other phagocytes to an immunosuppressive TME in glioblastoma and their prevalence within the tumor bulk is well known and has made macrophages an attractive therapeutic target for patients. For example, CSF/CSF-1R interactions have been shown to induce an immunosuppressive M2 phenotype in glioblastoma-associated microglia/macrophages, and blockade of CSF-1R improves survival in tumor burdened mice (Kennedy et al., 2013 [↗](#); Pyonteck et al., 2013 [↗](#)). However, our data and work from others suggests an equally important tumor-suppressive role for microglial/macrophages based on tumor stage and activation state. For example, serial transplantation of transformed mouse neural stem cells results in strong negative pressure from the immune system in syngeneic mice, leading to activation of tumor cell-intrinsic immune evasion mechanisms that support growth *in vivo* and *in vitro* (Gangoso et al., 2021 [↗](#)). In addition, recent spatial analyses of human glioblastoma tumor samples identified regions of activated (myeloperoxidase, MPO) macrophages that are associated with improved clinical outcomes, suggesting anti-tumorigenic roles for certain macrophage populations in patients (Karimi et al., 2023 [↗](#)). Our modeling further supports a strong anti-tumor role for pro-inflammatory phagocytes within the endogenous TME at initiation as well as following transplantation, which is considered an experimental surrogate for relapse potential. Our data also suggests that the pro-tumorigenic immune niche seen in later-stage gliomas is not already established at initiation stages. Given that the transition to mesenchymal-like glioblastoma is closely linked to immune cell infiltration and an immune-suppressive TME (Gangoso et al., 2021 [↗](#); Klemm et al., 2020 [↗](#)), it will be important to characterize these transitions to better understand anti-tumor mechanisms and their evolution during tumor progression, with the potential for therapeutically harnessing endogenous immune cell types and/or signaling pathways in the future. Altogether, these data suggest an anti-tumor growth but potentially relapse promoting role for therapeutic anti-macrophage approaches, highlighting the importance of future *in vivo* investigations into how inhibition of transitions and/or switching immunosuppressive microglia/macrophages to immune activating states may inhibit glioblastoma growth or prevent relapse for patients. Given the role for a complexity of cell types in inflammation, it will be interesting to further assess both the innate and adaptive immune systems. For example, based on common markers we observed less evidence for neutrophils within p53EPS, consistent with the relatively low presence of neutrophils seen in human glioblastoma (Friedmann-Morvinski and Hambardzumyan, 2023 [↗](#)). However, how these and other cell types change depending on specific genetic drivers and/or tumor stage should be highly accessible and translatable using our zebrafish model system given the broad availability of validated reporter transgenes, as well as the high level of genetic and physiological conservation with human.

Finally, it is well known that the type of glioma affects immune responses, implying that tumor-intrinsic factors shape the composition of the TME (Friebel et al., 2020 [↗](#); Gangoso et al., 2021 [↗](#); Klemm et al., 2020 [↗](#)). Therefore, phenotypic differences with respect to glioblastoma-TME interactions in patients and model systems are likely influenced by the genetic background and oncogenic driver events. Human AKT1 over-expression in zebrafish neural cells was previously shown to drive pre-neoplastic lesions in zebrafish larvae and was found to recruit macrophage and microglia populations through the Sdf1b-Cxcr4b signaling pathway (Chia et al., 2018 [↗](#)). In this model, loss of macrophages resulted in decreased oncogenic proliferation, suggesting tumor-promoting functions for macrophages at initiation stages. In contrast to AKT1-expressing pre-neoplastic neurons, we visualized that neutral red-positive phagocytes and *mpeg1.1:GFP*+ macrophages/microglia very closely associated with and/or engulfed mScarlet+ p53EPS tumor cellular puncta, suggesting an effort to clear cells and/or cellular debris in p53EPS. This anti-tumor role is further supported by our functional analyses showing increased p53EPS tumor initiation in the context of *Irf8* gene knock-out and decreased phagocytic cell development *in vivo* in zebrafish larvae. Given that effectively leveraging targeted therapies for glioblastoma will require a deep and individualized understanding of patient-specific tumor cell biology, it will be important to harness the flexibility of our mosaic oncogene over-expression approach and understand different tumor plus environmental factors in the context of different genetic drivers and/or modifiers identified in patients.

Methods

Zebrafish Husbandry and Care

Animals were raised in accordance with Canadian Council on Animal care (CCAC) guidelines and all experiments were approved under an Animal Use Protocol established with the Animal Care Committee at the Peter Gilgan Centre for Research and Learning (AUP #1000051391 and #1000064586). Previously described zebrafish strains including syngeneic CG1 (Mizgireuv and Revskoy, 2006 [↗](#)), CG1tp53^{del} (Ignatius et al., 2018 [↗](#)), Tg(*mpeg1.1:EGFP*) (Ellett et al., 2011 [↗](#)), and Tg(*tnfa:EGFP*) (Nguyen-Chi et al., 2015 [↗](#)) were used as indicated in the manuscript.

Preparation and injection of linearized DNA for tumorigenesis

her4.1:PI3KCA^{H1047R}, *her4.1:mScarlet*, *her4.1:EGFR^{ΔIII}*, *her4.1:KRAS^{G12D}*, *her4.1:EGFP* *her4.1:Cas9-2A-EGFP*, *gfap:EGFP*, and *gfap:mScarlet* transgene expression constructs were cloned using previously described *her4.1* sequence (Yeo et al., 2007 [↗](#)), *gfap* entry plasmid (Don et al., 2017 [↗](#)), standard Tol2 Gateway plasmids and protocols (Kwan et al., 2007 [↗](#)). *rag2:EGFP* and *mpeg1.1:EGFP* expression plasmids were previously described (Ellett et al., 2011 [↗](#); Langenau et al., 2007 [↗](#)). Circular *her4.1* expression plasmids were linearized using XhoI restriction enzyme (New England Biolabs, R0146S) while *gfap:EGFP* was linearized using ClaI (New England Biolabs, R0197S), according to manufacturer's protocol. Restriction enzymes were heat inactivated and linearized vector purified utilizing an EZ-10 Spin Column PCR Products Purification Kit (Bio Basic, BS364). Injection mixtures such as EPS (*her4.1:EGFR^{ΔIII}*, *her4.1:PI3KCA^{H1047R}*, *her4.1:mScarlet*) or KPG (*her4.1:KRAS*, *her4.1:PI3KCA^{H1047R}*, *her4.1:GFP*) were prepared at 2:1:1 molar ratios in 50% TE Buffer (Invitrogen, 12090-015) with KCl (final concentration 0.1M) and Phenol red (final concentration 5%, Sigma-Aldrich, P0290). CG1 or CG1tp53^{del} embryos were micro injected with 0.5-1nL of injection mixture at the one-cell stage and monitored for tumor development, starting at 10-15 days post fertilization (dpf).

Guide RNA Synthesis and CRISPR-Cas9-Mediated Gene Targeting

CRISPR/Cas9 sequence targets and guide RNA (gRNA) oligos were designed using CHOPCHOP (Labun et al., 2019 [↗](#)). Guide RNA were *in vitro* synthesized using the EnGen sgRNA Synthesis Kit (New England Biolabs, E3322V) according to manufacturer's recommendation. Cas9 protein with

NLS (PNA Bio, CP01-200) was resuspended in 20% glycerol/water to a concentration of 1mg/ml. Cas9/gRNA microinjection mix was prepared at a final concentration of 0.3mg/ml Cas9 + 30-50ng/ul of each gRNA. 0.5-1nL of injection mix and injected into zebrafish embryos of the indicated genotype, at the one-cell stage. CRISPR/Cas9 targeting efficiency was measured following PCR-based locus amplification, Sanger sequencing and Synthego ICE Analysis (ICE analysis, Synthego, ice.synthego.com) (Synthego, 2019). All gRNA sequences and PCR oligos are indicated in Supplementary Table 7.

Brain Dissection and Dissociation

Animals were euthanized with a lethal 300mg/L dose of tricaine (Sigma-Aldrich, E10521) ~10-20 minutes before dissection. Fish were decapitated posterior of the gills and the head transferred to sterile-filtered PBS (Wisent Inc., 311-010-CL). Using fine-tipped forceps, brains were carefully extracted from the skull and transferred to a micro centrifuge tube and kept in PBS at 28°C until further processing. 5-10 brains were transferred to 1ml of 28°C pre-warmed Accutase (Stemcell, 07920), followed by incubation at 28°C with gentle rocking for a total of ~50 minutes. Every 10 minutes, brains were mechanically dissociated with gentle pipetting 15-30 times using a 1mL filter tip. Dissociated tissues were passed through a 40um Cell Strainer (Corning, 352340) into a 50ml conical tube to achieve single cell suspensions. Strained cells were pelleted at approximately 1000 x g for 5 minutes at room temperature and resuspended in PBS.

Bulk Tumor Cell Transplantation

Cell suspensions were maintained at 28°C during the transplantation procedure. 2dpf CG1 strain syngeneic zebrafish larvae were injected with 1-2nL of cell suspension into the hindbrain ventricle, as previously described (Casey et al., 2017). Cells were injected alone or in combination with 1% total volume Clodrosomes (Encapsula Nano Sciences, CLD-8901), vehicle control liposomes, or 0.05ng/nL human CSF-1 recombinant protein (Cell Signaling Technology, 8929SC).

Immunohistochemistry and Fluorescent *in situ* Hybridization (FISH)

Zebrafish were fixed overnight in 4% PFA and stored in methanol at -20°C before paraffin embedding and sectioning at the Centre for Phenogenomics Pathology Core Facility. Animal sections were deparaffinized with 2x5 minute washes of Xylenes. Sections were rehydrated with sequential 2x10 minute washes in 100%, 90%, 70% ethanol before sequential rinsing with ddH₂O, 3% H₂O₂, and ddH₂O. Slides were then boiled in 1x Citrate Buffer (Sigma-Aldrich, C999) within a standard microwave and rinsed in ddH₂O, after cooling for 30 minutes. Sections were blocked with TBST/5% Normal Goat Serum for 1 hour at RT before overnight incubation with primary antibody diluted in Diluent CST (Cell Signaling Technology, 8112) at 4°C. The following day slides were washed once with 3x5 minutes with TBS, TBST, and TBS. Samples were then incubated with secondary antibody (Cell Signaling Technology, 8114) for half an hour at room temperature, followed by another 3x5 minute TBS/T wash. Standard DAB development (Thermo Scientific, 34002) was performed for 5-7 minutes, followed by water termination. Samples were Hematoxylin (VWR, 10143-146) stained with a 1:6 diluted solution for 3-5 seconds and rinsed 5x with tap water. Slides were dehydrated sequentially with 70%, 90%, 100% ethanol washes followed by 2x5 minute Xylene washes before being mounted with glass cover slips and Permount (Fisher Scientific, SP15-100). Primary antibodies raised against proliferating cell nuclear antigen (PCNA, Cell Signaling, D3H8P), phosphorylated ERK (Cell Signaling, #9101) and phosphorylated AKT (Cell Signaling, #9271) were used.

Whole mount FISH was performed on fixed larvae using *EGFP* and *irf7*-specific antisense probes (Molecular Instruments) and HCR protocol, according to manufacturer's recommendations.

Bulk RNASeq Library Preparation, Quantification and Differential Gene Expression Analysis

Three p53KPG, p53EPS, and control-injected brains (from non-tumor forming injected siblings) were harvested at 20-30 dpf and immediately placed into 1mL of TRIzol Reagent (Invitrogen, 15596026). mScarlet+ tumor cells were also sorted from bulk (non-fluorescent) tissue from pooled brains at the SickKids-UHN Flow Cytometry Core Facility on a Sony MA900 VBYS cell sorter, before pelleting and lysis in TRIzol Reagent. Total RNA was purified for all samples using a Monarch RNA Cleanup Kit (New England Biolabs, T2040L), according to manufacturer's recommendations. Sequence ready polyA-enriched libraries were prepared using the NEB Ultra II Directional mRNA prep kit for Illumina (NEB, E7760), according to manufacturer's recommendations. Single-end 150bp sequencing at a targeted depth of ~30-60 million reads/sample was performed using an Illumina NovaSeq S1 flowcell, at the Centre for Applied Genomics (TCAG). Raw .fastq data was processed using Salmon quantification of transcripts for each sample. A "decoy-aware" index was built with the *Danio rerio* transcriptome and genome using the GRCz11 assembly with a k-mers length of 23. Samples were then quantified with the following arguments: -r, --seqBias, --mp -3, --validateMappings, --rangeFactorizationBins 4. Sequencing data is available from GEO under accession code GSE246295.

Human Ortholog Conversion and GSEA

Normalized counts output from DESeq2 for the above processed data were utilized for GSEA analysis without any further trimming or processing, as recommend by the GSEA user guide (Mootha et al., 2003 [DOI](#); Subramanian et al., 2005 [DOI](#)). Zebrafish transcripts were assigned known or high-confidence Human orthologs using Ensembl BioMart, as previously described (Demirci et al., 2022 [DOI](#)). Bulk sequencing data from *her4.1:KRAS^{G12D}* or *her4.1:EGFR^{viii}* driven tumors, or sorted cells derived from such, were compared with non-tumour control brains under default conditions (1000 permutations, gene set). Expression signatures were compared against published glioblastoma (McLendon et al., 2008 [DOI](#); Wang et al., 2017 [DOI](#)), medulloblastoma (Cavalli et al., 2017 [DOI](#)), Hallmark and/or C7: immunologic signature gene sets (Liberzon et al., 2011 [DOI](#)), as indicated in the manuscript.

Neutral Red Staining and Quantification

Zebrafish were treated with Neutral Red, as previously described (Shiau et al., 2013 [DOI](#)). Animals were then anesthetized in Tricaine and oriented on their sides in 3% methylcellulose before an image stack of approximately 120-150um depth was taken beginning from the surface of the otic vesicle into the fish. Neutral Red stained foci were counted using Zen Lite Software (version 3.3) in an area bounded by the posterior edge of the eye and the posterior edge of the otic vesicle for each fish.

Image Acquisition

Fluorescence and bright field image acquisition of whole animals and dissected tumor-burdened brains was performed using a Zeiss Axiozoom V16 macroscope. High resolution confocal microscopy was performed using a Nikon 1AR confocal microscope, with image processing and Z-stack compression performed using ImageJ software. IHC slides were imaged on a Panoramic Flash II.

RT-qPCR and Analysis

Total RNA was extracted from tumor-burdened brains using TRIzol reagent, as described above, and reverse transcribed using the High-Capacity DNA Reverse Transcriptase Kit (Thermo Fisher, 4368814), as per manufacturer's recommendations. RT-qPCR was performed using the SYBR Green I Master kit (Roche, 04887352001) following the manufacturer's recommended protocol and a Bio-

Rad CFX96 qPCR Real-Time PCR Module with C1000 Touch Thermal Cycler Unit. All primers used to amplify genes of interest are indicated in Supplementary Table 7. At least 5 brain samples were pooled for each sample, with gene expression normalized to *18s*. Where indicated, gene expression was normalized to *mScarlet* expression to control for differences in overall tumor size.

Data Availability

All sequencing data is deposited in NCBI GEO under accession code GSE246295: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE246295> .

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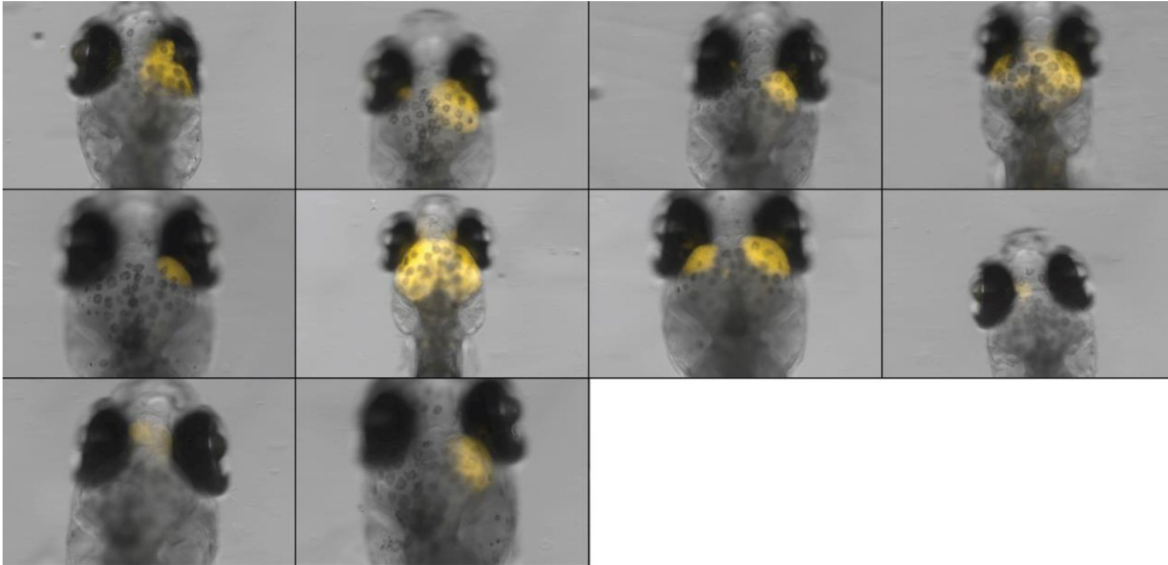
Supplementary Figures

Supplementary Movie 1. Time-lapse confocal images of p53EPS brain at 12 dpf in transgenic Tg(*mpeg1.1*:EGFP) background.

Supplementary Movie 2. Time-lapse confocal images of individual *her4.1*:mScarlet+ p53EPS and *mpeg1.1*:EGFP+ cells at 12 dpf.

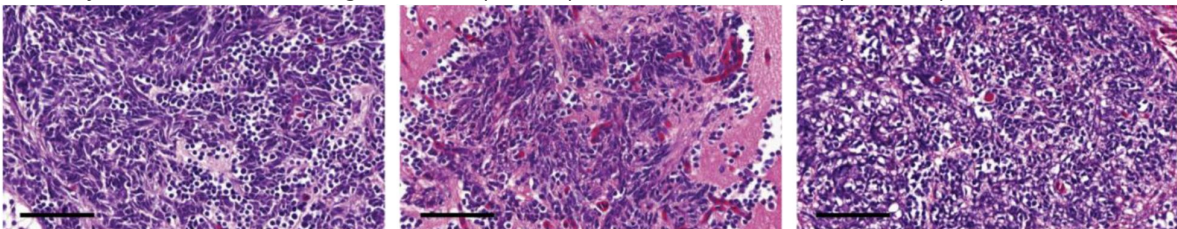
Supplementary Figure 1.

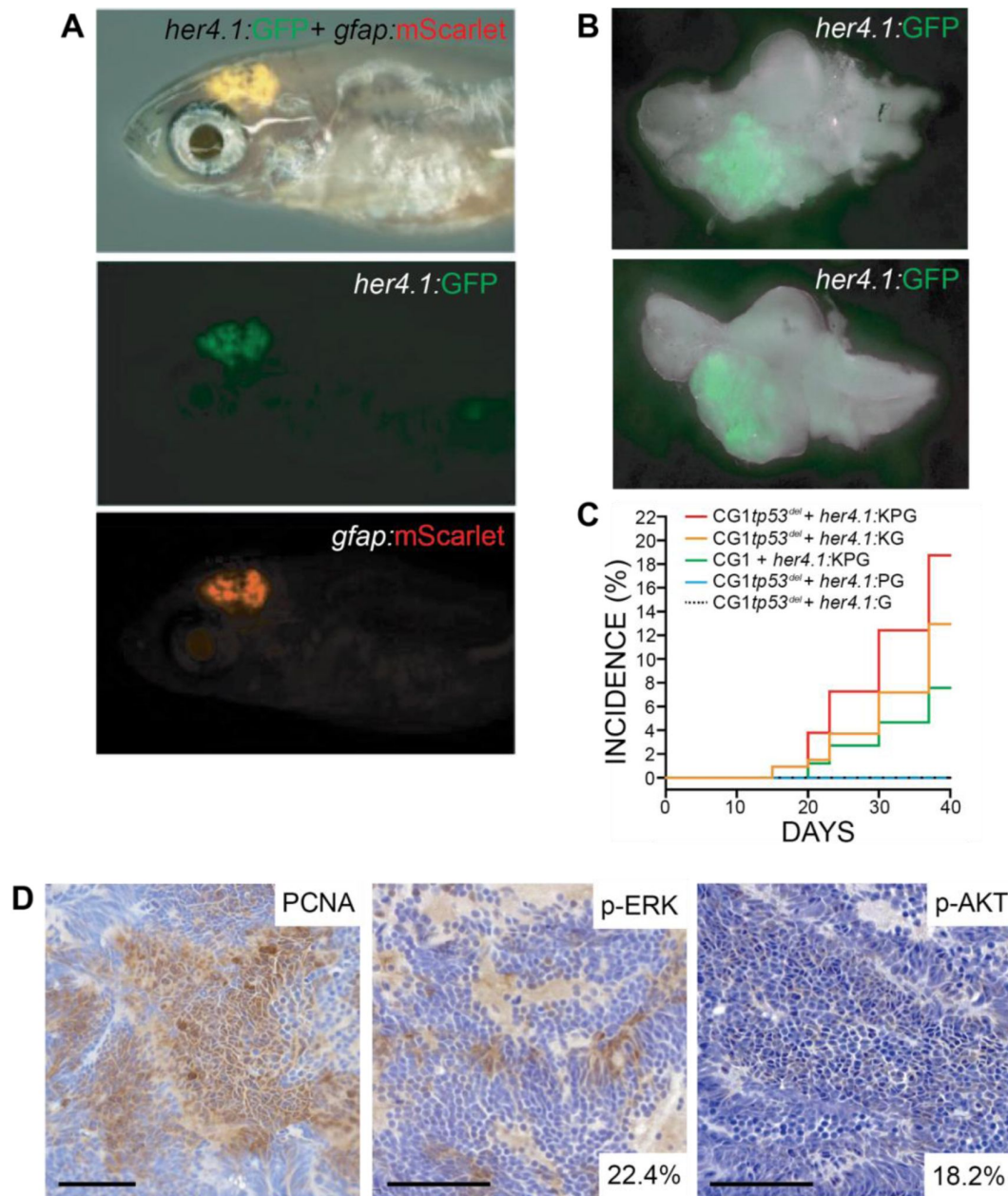
Inter-tumoral heterogeneity in p53EPS induced tumors. Sample screens of p53EPS injected fish showing distinct tumor initiation sites and varying degrees of mScarlet-fluorescent intensity. Tumors predominantly arise in the optic tectum/mesencephalon region with a subset appearing in the telencephalon/diencephalon region. Distinct tumors did not arise in the cerebellum/rhombencephalon regions over the course of our experiments.



Supplementary Figure 2.

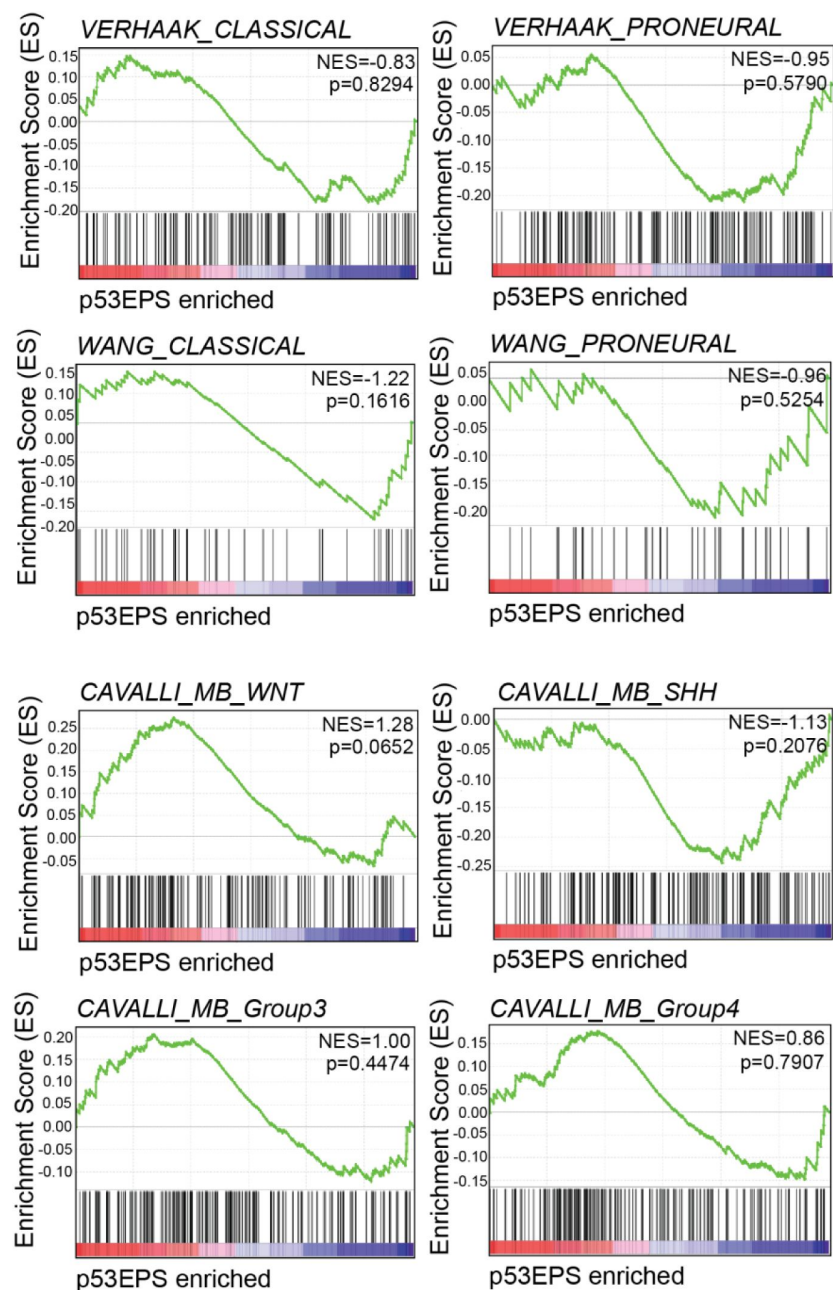
Hematoxylin and eosin (H&E) staining of three independent p53EPS tumors. Scale bars represent 50µm.





Supplementary Figure 3.

(A) *her4.1:mScarlet* and *gfap:GFP* expression in the anterior CNS of 30dpf syngeneic (CG1) *tp53*^{-/-} zebrafish injected at the one-cell stage with linearized *her4.1:KRAS*^{G12D} + *her4.1:PI3KCA*^{H1047R} + *her4.1:GFP* + *gfap:mScarlet* transgenes. **(B)** Whole brain dissected from mosaic-injected *her4.1:KRAS*^{G12D} + *her4.1:PI3KCA*^{H1047R} + *her4.1:GFP* zebrafish at 30dpf. **(C)** Cumulative frequencies of GFP+ brain lesions in syngeneic *tp53*^{-/-} mutant (CG1*tp53*^{-/-}) and wild-type (CG1) zebrafish injected at the one cell stage with *her4.1:KRAS*^{G12D} (K), *her4.1:PI3KCA*^{H1047R} (P), and/or *her4.1:GFP* (G). **(D)** Immunohistochemical staining of proliferating cell nuclear antigen (PCNA), phosphorylated ERK (p-ERK), and phosphorylated AKT (p-AKT) on *her4.1:KRAS*^{G12D} + *her4.1:PI3KCA*^{H1047R} + *her4.1:GFP* tumor section. Quantification of p-ERK and p-AKT positive cells within total field. Scale bars represent 50μm.

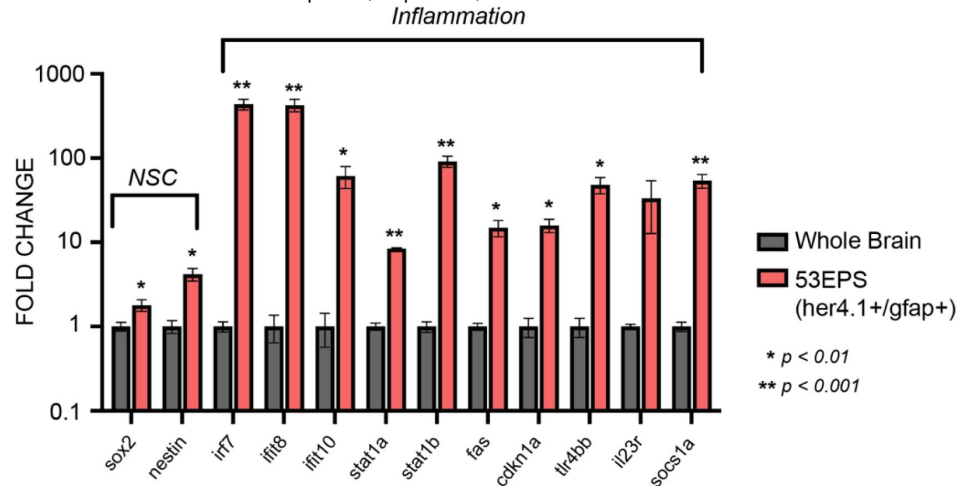


Supplementary Figure 4.

Gene set enrichment analysis (GSEA) plots of published gene signatures for alternative molecular subtypes of human glioblastoma and medulloblastoma for genes differentially regulated in p53EPS compared to control injected brains (Cavalli et al., 2017 [DOI](#); McLendon et al., 2008 [DOI](#); Wang et al., 2017 [DOI](#)). Normalized Enrichment Scores (NES) and nominal p-values are indicated.

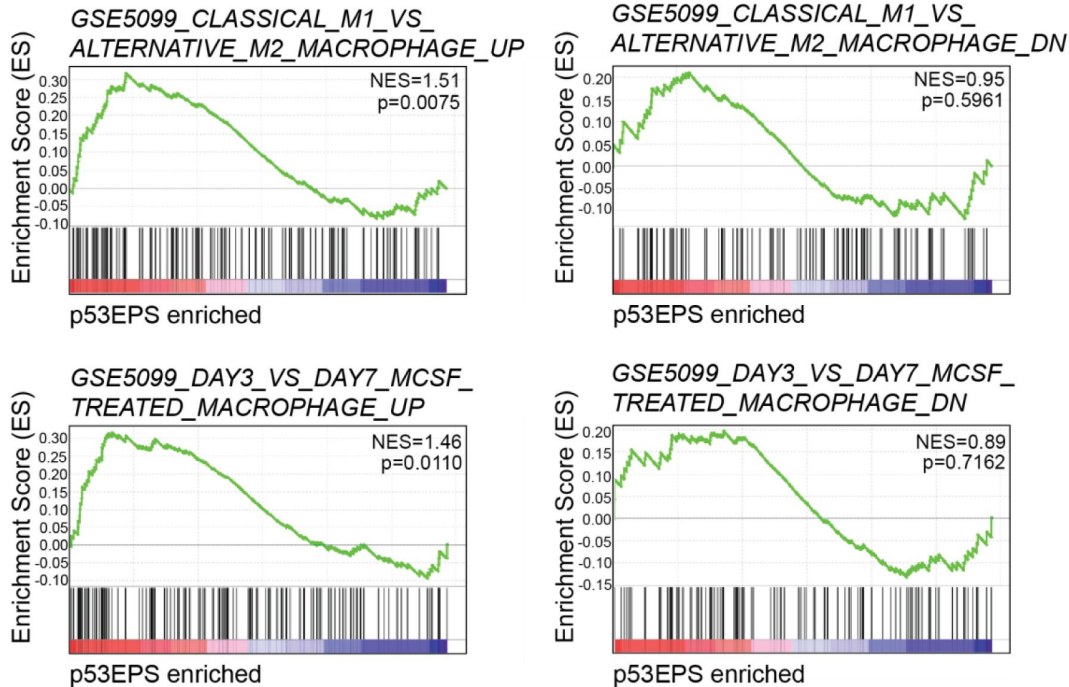
Supplementary Figure 5.

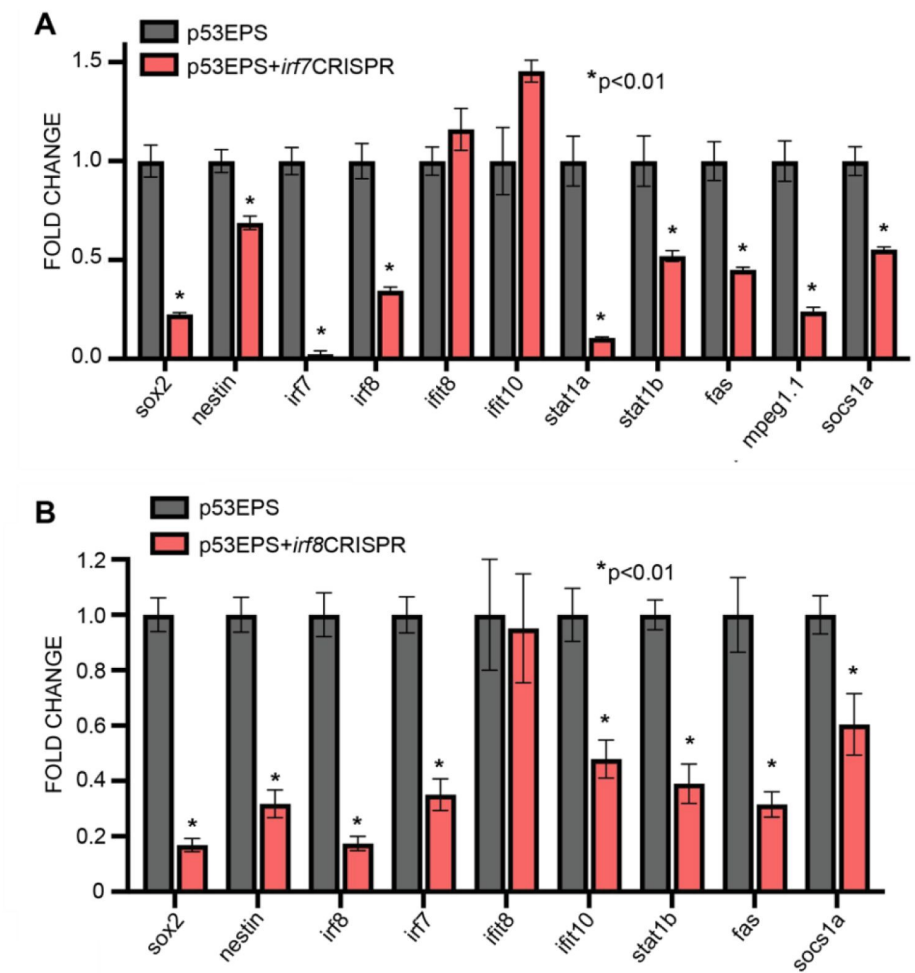
Quantitative real-time PCR analysis of neural stem cell (NSC) genes and genes associated with inflammatory gene expression signatures identified using bulk RNA sequencing. Gene expression in pooled *her4.1:mScarlet+/gfap:GFP+* FACS-sorted cells relative to non-tumor control brain tissue. * $p < 0.01$, ** $p < 0.001$, Student's t-test.



Supplementary Figure 6.

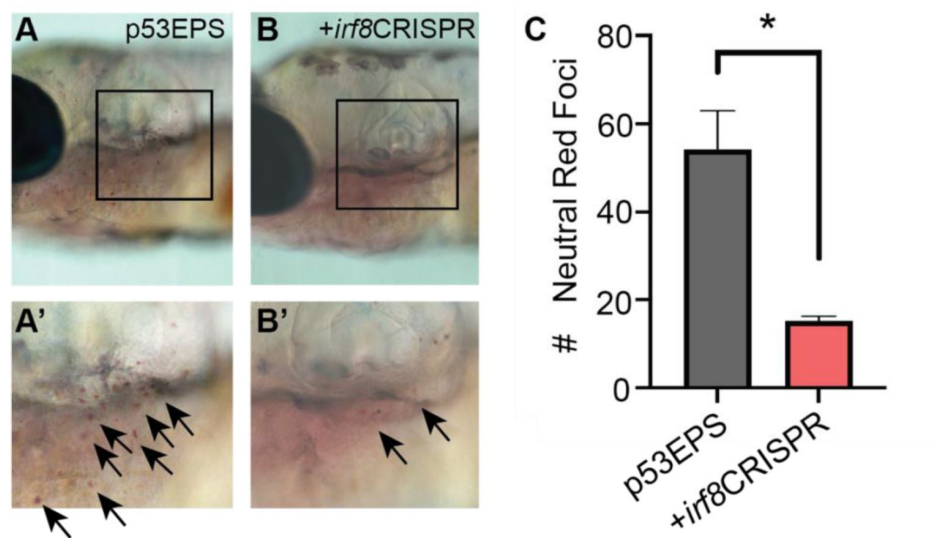
Gene set enrichment analysis (GSEA) plots of established gene signatures for classical M1 polarized macrophages (Classical_M1_VS_Alternative_M2_Macrophage_UP), compared to alternative M2 macrophages (Classical_M1_VS_Alternative_M2_Macrophage_DN). GSEA plots of macrophage-specific gene expression at early (DAY3 UP) and late (DAY3 Down) time points of stimulation with macrophage colony stimulating factor (MCSF). Normalized Enrichment Scores (NES) and nominal p-values are indicated.





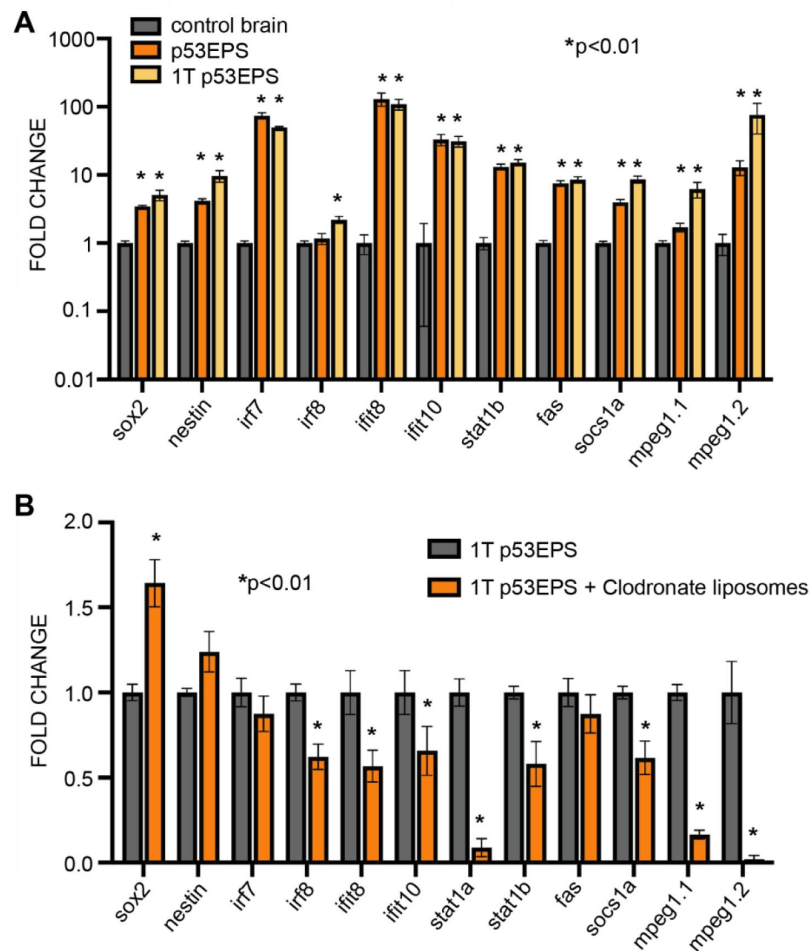
Supplementary Figure 7.

Quantitative real-time PCR analysis of *irf7* CRISPR/Cas9-injected (A) and *irf8* CRISPR/Cas9-injected p53EPS (B). Gene expression is represented relative to control p53EPS tumor-burdened whole brains. At least 5 tumor-burdened brains were pooled for each cohort and expression was normalized to mScarlet mRNA expression to account for differences in tumor size. *p<0.01, Student's t-test.



Supplementary Figure 8.

(A,B) Images of neutral red staining in control (A) and *irf8* CRISPR/Cas9-injected (B) zebrafish larvae at 8 dpf, A' and B' highlight boxed regions. Black arrows highlight examples of neutral red foci. **(C)** Quantification of neutral red foci. * $p < 0.01$, $n = 4$ fish for each cohort, Student's t-test.



Supplementary Figure 9.

(A) Quantitative real-time PCR analysis of neural stem cell (NSC) genes and inflammation genes in non-tumor control brains, pooled p53EPS-burdened brains and pooled brains engrafted with p53EPS tumor cells at 20 dpf. *p<0.01 compared to non-tumor control whole brain tissue, Student's t-test. **(B)** Quantitative real-time PCR analysis of neural stem cell (NSC) genes and inflammation genes at 20 dpf in primary transplanted (1T) p53EPS-burdened brains injected with vehicle control liposomes (1T p53EPS) and Clodrosomes at 2 dpf. At least 5 tumor-burdened brains were pooled for each cohort and expression was normalized to mScarlet mRNA expression to account for differences in tumor size. *p<0.01 compared to 1T p53EPS, Student's t-test.

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Editors

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Reviewer #1 (Public Review):

Summary:

The authors have developed a zebrafish model of glioblastoma and characterized this, with a particular focus on the role of recruited myeloid cells in the tumours. Microglia/macrophages in the tumours are proposed to have an inflammatory phenotype and are engaged in phagocytosis. Knockout of *Irf7* and *Irf8* genes enhanced tumour initiation. Depleting mature myeloid cell types with clodronate also enhanced tumour initiation. It is proposed that in early stage tumours, microglia/macrophages have tumour suppressive activity.

Strengths:

The authors have generated a novel glioblastoma model in zebrafish. Two key strengths of the zebrafish model are that early stage tumours can be studied and *in vivo* visualization can be readily performed. The authors show video of microglia/macrophages adopting the amoeboid phenotype in tumours (as is observed in human tumours) and engaging in phagocytosis. Video 1 was very impressive in my opinion and shows the model is a very useful tool to study microglia/macrophage:glioblastoma cell interactions. The *irf7/irf8* knockdown and the clodronate experiments are consistent with a role for mature myeloid cells in suppressing tumour initiation, suggesting that the model may also be very valuable in understanding immune surveillance in glioblastoma initiation.

Weaknesses:

EGFRvIII is mainly associated with the classical subtype, so the mesenchymal subtype might be unexpected here. This could be commented on. Some more histologic characterization of the tumours would be helpful. Are they invasive, do larger tumours show necrosis and microvascular proliferation? This would help with understanding the full potential of the new model. Current thinking in established human glioblastoma is that the M1/M2 designations for macrophages are not relevant, with microglia macrophage populations showing a mixture of pre- and anti-inflammatory features. Ideally there would be a much more detailed characterization of the intratumoral microglia/macrophage population here, as single markers can't be relied upon. Phagocytosis could have antitumour effects through removal of live cancer cells, or could be cancer promoting if apoptotic cancer cells are being rapidly cleared with concomitant activation of an immunosuppressive phenotype in the phagocytes (i.e. efferocytosis). It may be possible to distinguish between these two types of phagocytosis experimentally. Do the *irf7/8* and clodronate experiments distinguish between effects on microglia/macrophages and dendritic cells?

Update: The more detailed description of the tumour histology is very interesting and the authors have addressed my previous concerns nicely.

<https://doi.org/10.7554/eLife.93077.2.sa1>

Reviewer #2 (Public Review):

Summary:

Glioblastoma is a common primary brain cancer, that is difficult to treat and has a low survival rate. The lack of genetically tractable and immunocompetent vertebrate animal model has prevented discovery of new therapeutic targets and limited efforts for screening of pharmaceutical agents for the treatment of the disease. Here Weiss et al., express oncogenic variants frequently observed in human glioblastoma within zebrafish lacking the tumor suppressor TP53 to generate a patient-relevant *in vivo* model. The authors demonstrate that loss of TP53 and overexpression of EGFR, PI3KCA, and mScarlet (p53EPS) in neural progenitors and radial glia leads to visible fluorescent brain lesions in live zebrafish. The authors performed RNA expression analysis that uncovered a molecular signature consistent with human mesenchymal glioblastoma and identified gene expression patterns associated

with inflammation. Live imaging revealed high levels of immune cell infiltration and associations between microglia/macrophages and tumor cells. To define functional roles for regulators of inflammation on specific immune-related responses during tumorigenesis, transient CRISPR/Cas9 gene targeting was used to disrupt interferon regulator factor proteins and showed Inflammation-associated *irf7* and *irf8* are required to inhibit p53EPS tumor formation. Further, experiments to deplete the macrophages using clodronate liposomes suggest that macrophages contribute to the suppression of tumor engraftment following transplantation. The authors' conclusions are supported by the data and the experiments are thoroughly controlled throughout. Taken together, these results provide new insights into the regulation of glioblastoma initiation and growth by the surrounding microenvironment and provide a novel in vivo platform for the discovery of new molecular mechanisms and testing of therapeutics.

Strengths/Weaknesses:

The authors convincingly show that co-injection of activated human EGFRviii, PI3KCAH1047R, and mScarlet into TP53 null zebrafish promotes formation of fluorescent brain lesions and glioblastoma-like tumor formation. The authors include histological characterization of the tumors, as well as quantifications of p-ERK and p-AKT staining to highlight increased activation of the MAPK/AKT signaling pathways in their tumor model.

The authors use a transplantation assay to further test the tumorigenic potential of dissociated cells from glial-derived tumors in the context of specific manipulations of the tumour microenvironment.

The authors nicely show high levels of immune cell infiltration and associations between microglia/macrophages and tumor cells. Quantification of the emergence of macrophages over time in relation to tumor initiation and growth is provided and supports the observations of tumor suppressive activity of the phagocytes. The authors also attempt to delineate if other leukocyte populations are involved and observe tumor formation without significant infiltration of neutrophils.

The authors provide evidence for key genetic regulators of the local microenvironment, showing increased p53EPS tumor initiation following *Irf7* gene knock-down and loss of *irf7* expression in the TME.

<https://doi.org/10.7554/eLife.93077.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1:

"EGFRvIII is mainly associated with the classical subtype, so the mesenchymal subtype might be unexpected here. This could be commented on."

We acknowledge that EGFRvIII is most often associated with the classical subtype of glioblastoma and agree that mesenchymal subtype classification may be unexpected given the use of *her4.1:EGFRvIII* as a driver in our model. We would like to highlight the fact that our brain tumors do also express certain markers associated with the classical subtype including neural precursor and neural stem cell markers like *sox2*, *ascl1b*, and *gli2* (Supplementary Fig 4, 5; Supplementary Table 1-3). However, our transcriptomic data was not found to significantly enrich for classical subtype gene expression, compared to normal brains. This could be due to a significant contribution of normal brain tissue to our analyses

(bulk tumor burdened brains were harvested for RNA sequencing), as well as the significant contribution of mesenchymal subtype signatures and/or inflammatory gene expression in our brain tumor-positive samples. Because signatures associated with inflammation consist of some of the most highly upregulated genes in our samples, this could potentially dilute out and/or lessen alternative subtype and/or signature gene expression. Importantly, it is now widely appreciated that patient tumors simultaneously consist of heterogeneous tumor cells reflecting multiple molecular subtypes (Couturier et al., 2020; Darmanis et al., 2017; Neftel et al., 2019), providing glioblastoma with a high level of phenotypic plasticity. We also demonstrate that the contribution of additional drivers not always present with EGFRvIII in patient glioblastoma enhances primary brain tumors *in vivo*. This result is consistent with more aggressive glioblastomas seen in patients with EGFRvIII variants and TP53 loss-of-function mutations (Ruano et al., 2009). It will therefore be interesting in the future to consider how single or multiple driver mutations contribute to subtype-specific gene expression in our model, as well as histopathology, relative to patients. We have included some of these discussion points to our revised manuscript.

“Some more histologic characterization of the tumors would be helpful. Are they invasive, do larger tumors show necrosis and microvascular proliferation? This would help with understanding the full potential of the new model.”

We have updated our manuscript to include more histopathological characterization and images (Supplementary Fig 2).

“Current thinking in established glioblastoma is that the M1/M2 designations for macrophages are not relevant, with microglia/macrophage populations showing a mixture of pre- and anti-inflammatory features. Ideally, there would be a much more detailed characterization of the intratumoral microglia/macrophage population here, as single markers can't be relied upon.”

We performed additional gene set enrichment analyses (GSEA) using our sequencing datasets and compared p53EPS gene expression to M1/M2 macrophage expression signatures and expression signatures from MCSF-stimulated macrophages at early and late (M2 polarized) time-points. From this analysis, we detected enrichment for markers of both pro- and anti-inflammatory features, however, with stronger and significant enrichment for gene expression signatures associated with classical pro-inflammatory M1 macrophages. We have included these GSEA plots and gene set enrichment lists as supplementary materials (Supplementary Fig 6, Supplementary Table 6). We also performed GSEA against a broad curated set of immunologic gene sets (C7: immunologic signature gene sets, Molecular Signatures Database, (Liberzon et al., 2011)) and have included the list of signatures and enrichment scores as a supplementary table (Supplementary Table 6).

“Phagocytosis could have anti-tumor effects through removal of live cancer cells or could be cancer-promoting if apoptotic cells are being rapidly cleared with concomitant activation of an immunosuppressive phenotype in the phagocytes (ie. efferocytosis).”

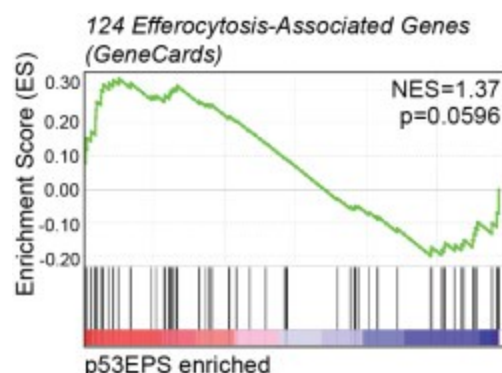
We looked at efferocytosis-associated gene expression in our sequencing dataset (124 “efferocytosis” genes, GeneCards), and while we detected upregulation of certain genes associated with efferocytosis in p53EPS brains, we did not detect significant enrichment for the entire gene set. Furthermore, we did not detect up-regulation of key efferocytosis receptors including Axl and Tyro3 (Supplementary Table 1, 2), compared to normal brains. While efferocytosis may contribute to tumor growth and evolution, this GSEA combined with our functional data supporting an inhibitory role for phagocytes in p53EPS tumor initiation and engraftment following transplantation (Fig 4, Fig 5, Supplementary Fig 7), suggests that efferocytosis is not a major driver of tumor formation in our model. However, how efferocytosis affects tumor progression in our model and/or relapse following therapy will be

an interesting feature to explore in the future using temporal manipulations of phagocytes and/or treatments with chemical inhibitors.

Author response image 1.

Gene Set Enrichment Analysis (GSEA) for efferocytosis-associated gene expression (124 “efferocytosis” genes in GeneCards) in tp53EPS tumor brains, compared to normal zebrafish brains.

Normalized enrichment score (NES) and p-value are indicated.



“Do the irf7/8 and chlodronate experiments distinguish between effects on microglia/macrophages and dendritic cells?”

In addition to microglia/macrophages, the IRF8 transcription factor has been shown to control survival and function of dendritic cells (Sichien et al., 2016). Chlodronate treatments are also used to deplete both macrophages and dendritic cells *in vivo*. Therefore, we cannot distinguish the effects of these manipulations in our experiments and have updated our manuscript throughout to reflect this.

Reviewer #2:

“The authors state that oncogenic MAPK/AKT pathway activation drives glial-derived tumor formation. It would be important to include a wild-type or uninjected control for the pERK and pAKT staining shown in Fig1 I-K to aid in the interpretation of these results. Likewise, quantification of the pERK and pAKT staining would be useful to demonstrate the increase over WT, and would also serve to facilitate comparison with the similar staining in the KPG model (Supp Fig 2D).”

We have updated Fig 1 and Supplementary Fig 3D (formerly Fig 2D), to include histology from tumor-free uninjected control animals, as well as quantifications of p-ERK and p-AKT staining to highlight increased MAPK/AKT signaling pathway activation in our tumor model.

“The authors use a transplantation assay to further test the tumorigenic potential of dissociated cells from glial-derived tumors. Listing the percentage of transplants that generate fluorescent tumor would be helpful to fully interpret these data. Additionally, it was not clear based on the description in the results section that the transplantation assay was an “experimental surrogate” to model the relapse potential of the tumor cell. This is first mentioned in the discussion. The authors may consider adding a sentence for clarity earlier in the manuscript as it helps the reader better understand the logic of the assay.”

We have clarified in the text the percentage of transplants that generated fluorescent tumor (1625%, n=3 independent screens). This is also represented in Fig 5C,D. We also added text when introducing the transplantation assay, explaining that transplantation is frequently used as an experimental surrogate to assess relapse potential, and that our objective was to assess tumor cell propagation in the context of specific manipulations within the TME.

"The authors nicely show high levels of immune cell infiltration and associations between microglia/macrophages and tumor cells. However, a quantification of the emergence of macrophages over time in relation to tumor initiation and growth would provide significant support to the observations of tumor suppressive activity of the phagocytes. Along these lines, the inclusion of a statement about when leukocytes emerge during normal development would be informative for those not familiar with the zebrafish model."

In zebrafish, microglia colonize the neural retina by 48 hpf, and the optic tectum by 84 hpf (Herbomel et al., 2001), prior to when we typically observe lesions in our p53EPS brains. To validate the emergence of microglia prior to tumor formation in p53EPS, we have now used live confocal imaging through the brains of uninjected control and p53EPS injected zebrafish at 5, 7 and 9 dpf. As expected, microglia were present throughout the cephalic region and in the brain at 5 dpf (120 hpf). At this stage, p53EPS injected zebrafish brains displayed mosaic cellular expression of *her4.1:mScarlet*; however, cells were sparse and diffuse, and no large intensely fluorescent tumor-like clusters were detected at this stage (n=12/12 tumor negative). At 7 dpf, microglia were observed in the brains of control and p53EPS zebrafish; however, at this stage we detected clusters of *her4.1:mScarlet*⁺ cells (n=5/9), indicative of tumor formation. Lesions were found to be surrounded and/or infiltrated by *_mpeg:_EGFP*⁺ microglia. Finally, at 9 dpf *her4.1:mScarlet*⁺ expression became highly specific to tumor lesions, and these lesions were associated with *_mpeg:_EGFP*⁺ microglia/macrophages (n=8/8 of tumor-positive zebrafish). These descriptions along with representative images has been added to Figure 3.

"From the data provided in Figure 4G and Supp Fig 7b, the authors suggest that "increased p53EPS tumor initiation following Irf gene knock-down is a consequence of irf7 and irf8 loss-of-function in the TME." Given the importance of the local microenvironment highlighted in this study, spatial information on the form of in situ hybridization to identify the relevant location of the expression change would be important to support this conclusion."

We performed fluorescent *in situ* hybridization (using HCR RNA-FISH, Molecular Instruments) on whole mount control and *irf7* CRISPR-injected p53EPG animals (*her4.1:EGFRvIII* + *her4.1:PI3KCAH1047R* + *her4.1:GFP*, GFP was used in this case because of probe availability).

Representative confocal projections through tumors, as well as single optical sections are presented and discussed in Figure 4, highlighting the location of *irf7* expression change following gene knock-down. We found significant *irf7* signal in and surrounding p53EPS tumors at early stages of tumor formation. This expression was reduced and/or lost following *irf7* CRISPR gene targeting, consistent with RT-PCR data (Supplementary Fig 7).

"The authors used neutral red staining that labels lysosomal-rich phagocytes to assess enrichment at the early stages of tumor initiation. The images in Figure 3 panel A should be labeled to denote the uninjected controls to aid in the interpretation of the data. In Supplemental Figure 6, the neutral red staining in the irf8 CRISPR-injected larvae looks to be increased, counter to the quantification. Can the authors comment if the image is perhaps not representative?"

We have updated Figure 3 and Supplementary Figure 6 to aid in the interpretation of our results. In Fig 3A, we used tumor-negative controls from our injected cohorts. This was done to control for exogenous transgene presence and/or over-expression prior to (or in the absence of) malignant transformation. In Supplementary Fig 6, our images are representative, but we have now used unprocessed images with arrowheads to highlight neutral-red positive foci for clarity. In our original manuscript the images contained software generated markers, which could have obscured and/or confused the neutral red staining we were trying to highlight.

Recommendations For the Authors:

Reviewer #1:

"The PI 3-kinase does a lot more than just activating mTOR and Akt – I would suggest modifying that sentence in the introduction."

We have adjusted text in the introduction to reflect the broad role for PI3K signaling.

Reviewer #2:

"In Supplemental Fig 1, it would be helpful for the authors to provide a co-stain, such as DAPI to label all nuclei, which would allow the reader to assess the morphology of the cells in the context of the surrounding tissue."

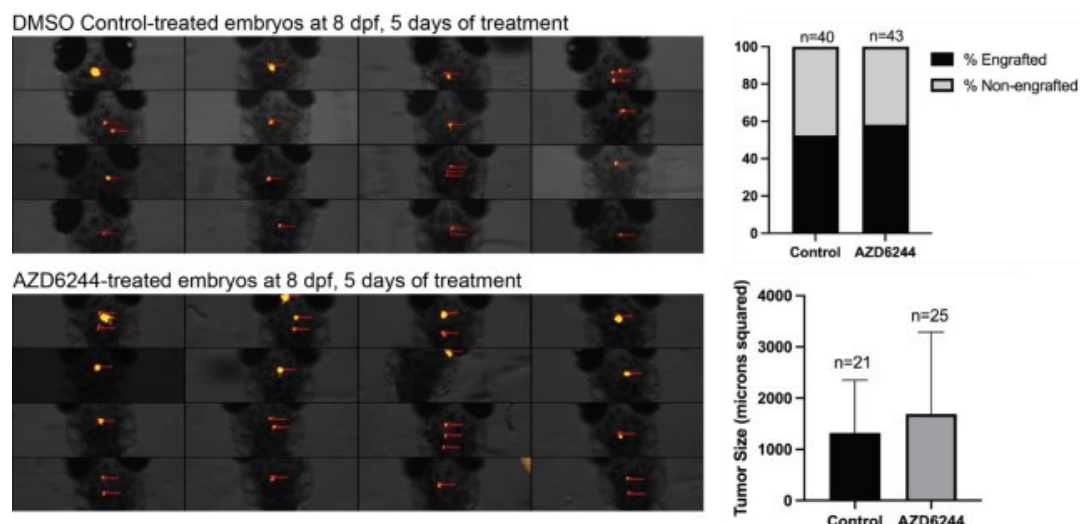
We have included brightfield images in Supplementary Fig 1, that together with *her4.1:mScarlet* fluorescence, should help readers assess tumor location and morphology in the context of surrounding tissue. Tumor cell morphology at high-resolution can be visualized in Fig 3, Movie 1 and Movie 2.

"The authors state that oncogenic MAPK/AKT pathway activation drives glial-derived tumor formation. The authors may consider testing if the addition of an inhibitor of MAPK signaling may prevent or decrease the formation of glial-derived tumors in this context to further support their results."

To further assess the role for MAPK activation, we decided to test the effect of 50uM AZD6244 MAPK inhibitor following transplantation of dissociated primary p53EPS cells into syngeneic CG1 strain zebrafish embryos, similar to as previously described (Modzelewska et al., 2016). Following 5 days of drug treatments, we did not detect significant differences in tumor engraftment or in tumor size between DMSO control and AZD6244-treated cohorts, suggesting that MAPK inhibition is not sufficient to prevent p53EPS engraftment and growth in our model. In the future, assessments of on-target drug effects, possible resistance mechanisms, and/or testing MAPK inhibitors in combination with other targeted agents including Akt and/or mTOR inhibitors (Edwards et al., 2006; McNeill et al., 2017; Schreck et al., 2020) will enhance our understanding of potential therapeutic strategies.

Author response image 2.

Dorsal views of 8 dpf zebrafish larvae engrafted with *her4.1:mScarlet+* p53EPS tumor cells following treatment from 3-8dpf with 0.1% DMSO (control) or 50uM AZD6244. Tumor cell injections were performed at 2 dpf into syngeneic CG1 strain embryos. The percentage of total animals with persisting engraftment following drug treatments, as well as tumor size (microns squared, quantified using Carl Zeiss ZEN software) are shown for control and AZD6244 treated larvae.



"Have the authors tested if EGFR and PI3KCA driven by other neural promoters produce similar results, or not? This would help support the specificity of her4.1 neural progenitors and glia as the cell of origin in this model."

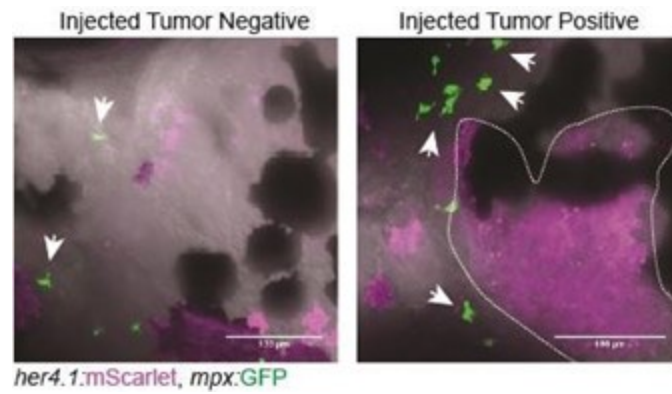
At this time, we have not tested other neural promoters. However, previous reports describe a zebrafish *zic4*-driven glioblastoma model with mesenchymal-like gene expression (Mayrhofer et al., 2017), supporting neural progenitors as a cell of origin. In the future it will be interesting to test *sox2*, *nestin*, and *gfap* promoters to further define and support *her4.1*-expressing neural progenitors and glia as the cell of origin in our model.

"Other leukocyte populations, such as neutrophils, can also respond to inflammatory cues. Can the authors comment if neutrophils are also observed in the TME?"

We performed initial assessments of neutrophils in the TME using our expression datasets as well as *her4.1:EGFRvIII* + *her4.1:PI3KCAH1047R* co-injection into Tg(*mpx:EGFP*) strain zebrafish. We observed tumor formation without significant infiltration of *mpx:EGFP*+ neutrophils. Future investigations will be important to assess differences in the contributions of different myeloid-derived lineages in the TME of p53EPS, as well as how heterogeneity may be altered depending on different oncogenic drivers and/or stage of tumor progression, as seen in human glioblastoma (Friedmann-Morvinski and Hambardzumyan, 2023). We have added text in the discussion section of our manuscript to indicate the possibility of neutrophils and/or other immune cell types contributing to p53EPS tumor biology.

Author response image 1.

Control-injected tumornegative and tumor-positive Tg(*mpx:EGFP*) zebrafish at 10 dpf. Tg(*mpx:EGFP*) strain embryos were injected at the one-cell stage with *her4.1:EGFRvIII* + *her4.1:PI3KCAH1047R* + *her4.1:mScarlet*.



"It is not clear if the transcriptomics data has been deposited in a publicly available database, such as the Gene Expression Omnibus (GEO). Sharing of these data would be a benefit to the field and facilitate use in other studies."

We have uploaded all transcriptomic data to GEO under accession GSE246295.

<https://doi.org/10.7554/eLife.93077.2.sa3>